Acknowledgements

[TODO: write at the end]

Contents

1 Introduction

A compound that enters a cell changes what that cell transcribes. Which genes move, by how much, and in which direction is the most direct readout available of what the compound actually does — not what it was designed to bind, but what the cell does in response. Reading that response is routine. Reading it for every combination that might matter is not. A compound behaves differently at different doses, and differently again depending on the genetic background of the cell receiving it, so the space to be explored is the product of three large sets rather than any one of them. Even the largest screens sample a small corner of it, and every combination left untested is a question left open.

This is the gap prediction is meant to close. If the transcriptional response to a treatment could be computed rather than measured, the untested corners of that space would become inspectable, and the experiments actually run could be chosen because they are informative rather than because they happened to fit on the plate. The task, stated plainly, is this: given the transcriptome of an untreated cell and the identity of a drug, predict the transcriptome that cell would have shown had the drug been applied.

The task is harder than it sounds, for a reason that is easy to state and impossible to engineer around. The ground truth is a counterfactual. Sequencing destroys the cell it reads, so the treated and untreated profiles of the same cell can never both be observed, and a prediction can only ever be checked against a different cell that received the drug. Everything downstream inherits this. It is why paired training data must be assembled by matching cells rather than by tracking them, why evaluation is usually framed at the level of populations rather than individuals, and why distinguishing a real drug effect from ordinary cell-to-cell variation requires care rather than optimism.

Assembling data at the scale this problem demands became possible only recently. Two decades of falling sequencing costs, rising throughput and increasingly clever sample multiplexing turned single-cell profiling from a delicate experiment into something that can be run industrially, and Tahoe-100M [1] is what the end of that trajectory looks like: roughly a hundred million single-cell transcriptomes spanning some 1,100 small molecules across 50 cancer cell lines. Every compound and every cell line appears not as a summary statistic but as a cloud of individual cells.

That resolution changes which questions can be asked. The obvious one is conditional — what does this compound do to this cellular context, rather than what does it do on average across contexts that may have little in common. The less obvious question is whether the first has an answer. Because many conditions were run more than once, it becomes possible to measure how much of an apparent drug effect reproduces across replicates and how much is simply the variability of the assay. That second measurement turns out to matter as much as the first, and a good deal of what follows rests on the observation that standard evaluations quietly confuse the two.

The other development behind this thesis has nothing to do with cells. Over the same period, pretrained transformer language models became unreasonably good and unreasonably general, and the engineering poured into them — architectures, training recipes, scaling laws, the entire surrounding toolchain — came to exceed what any single scientific field could plausibly assemble for itself. That created a standing temptation: if a problem can be made to look like text, it inherits all of that work for free.

Cell2Sentence [2] takes the temptation about as literally as it can be taken. Rank the genes in a cell from most to least expressed, write their names out in that order, and the cell becomes a sentence — ordinary text that an ordinary language model can be fine-tuned on. The encoding is unsentimental about biology, discarding how much each gene is expressed and keeping only the ordering, on the argument that rank retains enough of the signal for the rest to be reconstructed. In exchange it requires no new architecture and no new training machinery, only cells written in a format that existing models already consume.

C2S-Scale [3] develops that idea into a family of models. Larger backbones are pre-trained on over 50 million cells drawn from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas, and then fine-tuned across a spread of tasks: predicting cell type and tissue, generating cells conditioned on a label, captioning clusters, answering questions about a dataset in natural language, and predicting perturbation responses. That last task is the one this thesis builds on, and it is worth being specific about its scope. Their perturbation work covers cytokine stimulation in peripheral blood cells, and, for small molecules, bulk L1000 profiles and the SciPlex3 panel. The single-cell results are scored as distributions — how closely the generated population for a condition resembles the observed one — rather than condition by condition. The checkpoint used throughout this thesis, `C2S-Scale-Pythia-1b-pt`, is the pretrained model from that family: atlas data only, with no drug-treated cells among it.

What we do is extend that line of work to chemical perturbation at the scale Tahoe-100M now makes available, and evaluate it as a prediction problem in the strict sense — condition by condition, against controls chosen so that a model which ignores the drug cannot pass. The question is whether a cell-sentence model fine-tuned on this atlas learns what a particular compound does to a particular cell.

Two things are worth keeping straight about what the model has actually seen, since the scale of the atlas invites an easy overstatement. Its pretraining is large but drug-free: tens of millions of cells, none of them treated with anything. And the fine-tuning does not consume Tahoe entire — it draws a sample of a few hundred thousand treated cells from it, which is ample for the task but is not a hundred million. Whatever the model learns about drugs, it learns during fine-tuning, from that sample.

It does learn something, and the something is the difficulty. The model acquires a good account of what a treated cell looks like *in general* — the shared shape that perturbed transcriptomes have in common, which is substantial and largely the same whichever compound produced it. What it does not acquire is the part that distinguishes one compound from another. Change the drug named in the prompt while holding the input cell fixed, and the prediction barely moves; on the calibrated metric the model is statistically indistinguishable from a baseline that copies the untreated cell and is told nothing about any drug at all.

A second comparison is worth separating from that one, because it is weaker but more informative. A per-drug average, estimated from the same compound in *other* cell lines, also outperforms the model — but this is not a naive baseline. It is very nearly the best any method can do at the part of the problem it addresses, scoring 0.963 against a replicate ceiling of 0.968, and being close to it would be a real achievement rather than a low bar. What it cannot do is condition on the cell in front of it, which is precisely why the comparison is useful: it separates the drug's average effect from the part that depends on context. Section 3.9 measures that split, and finds the second part worth roughly half the drug-specific signal — reached by the average, by the model, and by the literature alike, which is to say not at all.

A negative result earns its place only if it can be made specific, and that is the organising principle of what follows. We establish first that the failure is real rather than an artifact of how this field measures success — a distinction less pedantic than it sounds, since the standard metric can be saturated by a predictor that knows nothing about drugs. We then locate where in the pipeline the failure originates, show that the location implies a repair, apply it, and finally bound how much of the problem any method of this kind could recover even if that repair were perfect. The thesis therefore ends not in closure but in a number: a measured fraction of the drug-specific signal claimed by nothing in the literature, by no baseline we could construct, and by none of our own models.

Four questions carry that argument, in order:

1. **Is the evaluation trustworthy?** Before asking whether a model is any good, one has to ask whether the metric could tell. Can a predictor carrying no drug information score well on the measures the field has standardised on, and if so, which measures survive?

2. **Does the model use the drug at all?** Holding the input cell fixed and changing only the compound it is told about, does the prediction move — and does it move in the direction the real response moved?

3. **If not, where does the failure live?** Is the drug missing from the data, missing from the model's internal representation, or present in that representation but ignored by the process that writes the output? Each answer implies a different repair, and only one of them is cheap.

4. **How much is recoverable at all?** A per-drug average is a strong baseline. How much of the drug-specific signal does it capture, how much is genuinely conditional on cellular context, and how much does any method — ours or anyone else's — actually reach?

The contributions follow the same order. We audit the calibration of the standard evaluation and show that it can be saturated by a predictor carrying no drug information whatsoever, which bears on any work built from the same family of metrics. We establish drug-blindness on the one metric that survives that audit, using several independent instruments so that no single measurement carries the claim alone. We identify the target encoding — not the data, the architecture, or the training objective — as the binding constraint, and derive from that diagnosis a re-encoding of the prediction target which produces the first measurable drug effect in this project and generalises to drug-and-cell-line pairings never seen during training. And we separate the per-drug main effect from the drug-by-cell-line interaction, showing that the interaction accounts for roughly half the drug-specific signal and is reached by neither the lookup baseline nor the model.

The thesis proceeds as follows. Section 2 places the work against single-cell foundation models, the perturbation-prediction literature, and an unsettled dispute within it about whether deep models beat trivial baselines at all — a dispute this thesis adjudicates in one corner, using the instruments of the side it ultimately argues against. Section 3 is the investigation itself, told in the order it happened: because nearly every instrument in it was built in response to the measurement that came before, the data, the methods and the findings are presented as one continuous account, with each definition given where it is first needed. Section 4 sets out the scope honestly, separating what is closed by measurement from what is merely untested, and Section 5 concludes.

2 Literature review

2.1 From text foundation models to single-cell foundation models

The models this thesis builds on descend from a template established in natural language processing. A transformer is pretrained on a very large unlabelled text corpus with a single objective — predict the next token — and the resulting network is then adapted to specific tasks by fine-tuning or prompting. The GPT line of models made the case that this recipe scales: capability on tasks never explicitly trained for improves steadily with model size, data and compute, so a single pretrained backbone can serve as a general substrate rather than a task-specific tool. What made the recipe portable is that it assumes almost nothing about language. It requires only that the data be expressible as sequences of discrete tokens, and that some tokens be predictable from others.

That portability is what invited the transfer into biology. If a cell can be written as a sequence, the same machinery applies unchanged. scGPT [4] and Geneformer [5] both take this route, treating a cell's expression profile as a sequence of gene tokens and pretraining on tens of millions of cells before transferring to classification, integration and perturbation tasks. The ambition is usually stated as some version of the *virtual cell* [6]: a model whose simulated responses to intervention are trustworthy enough to substitute for an experiment.

The interesting variation among these models is not architectural but representational. A cell is a vector of expression values over some twenty thousand genes; a transformer consumes a sequence of discrete tokens. Something must be discretised or discarded in between, and what gets discarded bounds what the model can learn. The approach this thesis uses makes that choice unusually explicit.

2.2 Cell sentences, and the model this thesis fine-tunes

Cell2Sentence [2] takes the most literal route available: rank a cell's genes by expression and write their symbols out in descending order. The cell becomes a *cell sentence*, which is text in the ordinary sense — a string that a tokeniser will happily consume:

```
MALAT1 ACTB TMSB4X B2M EEF1A1 RPL13A FTL TPT1 RPS27 GAPDH ...
```

Nothing marks this as biological. The information is carried entirely by which symbols appear and in what order, and a language model fine-tuned on such strings is doing exactly what it does on prose: predicting the next token given the previous ones.

The encoding keeps rank and discards magnitude. C2S defends the loss with a reconstruction argument: expression can be recovered from rank through a fitted log-linear relationship $e_i = a_d \log(r_i) + b_d$, which achieves $R^2 = 0.815$ against the original values on peripheral blood data. On that basis they argue rank encodes expression reversibly enough across datasets for the discarded magnitude not to matter for downstream use. It is worth noting what kind of criterion this is — reconstruction fidelity is a statement about *absolute* accuracy, how close the recovered profile is to the true one, which is not automatically the same as preserving whatever distinguishes one experimental condition from another.

Understanding what the model does with such a sentence requires a little of the underlying architecture, because several of its properties matter later. C2S-Scale builds on decoder-only transformers of the GPT family, and the checkpoint used here is built on Pythia [7], a suite of such models released with open weights and training data. A decoder-only transformer factorises the probability of a sequence into a product of next-token predictions,

$$p(x_1, \ldots, x_T) \;=\; \prod_{t=1}^{T} p(x_t \mid x_{<t}) \,, \tag{1}$$

and is trained by minimising the cross-entropy between each predicted distribution and the token that actually occurred. Each token is first mapped to a vector by an embedding matrix, and those vectors form what is usually called the *residual stream*: a running representation, one vector per position, which every layer reads from and writes back into additively. A layer comprises multi-head self-attention, which lets each position gather information from earlier positions and is *causally masked* so that position t attends only to positions $\leq t$, followed by a position-wise feed-forward network. Both are wrapped in normalisation and added back to the stream rather than replacing it. After the final layer an unembedding matrix projects the stream to one score per vocabulary entry, and a softmax turns those scores into the next-token distribution.

Three properties of that design are worth noting now, since later sections depend on them. The residual stream is additive and shared, so a representation can be read at a chosen depth, modified, and the remaining computation allowed to proceed on the modified value — which is what makes interpretability interventions of the kind used in Section 3.6 possible at all. The training loss is a sum of per-token cross-entropies that weights every position equally, so a target whose tokens are largely shared between conditions gives any individual condition only a small share of the gradient. And conditioning is mediated entirely by attention across a causal mask, so information supplied early in a prompt must be reached from every later position at which it is needed.

C2S-Scale [3] develops Cell2Sentence into a family of models on this architecture, spanning backbones from Pythia at 410 million and 1 billion parameters up to Gemma-2 at 27 billion. Pretraining uses over 50 million human and mouse cells drawn from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas. The pretrained models are then fine-tuned across a range of tasks: predicting cell type and tissue, generating cells conditioned on a label, captioning clusters, answering natural-language questions about a dataset, and predicting perturbation responses.

The perturbation task is the one this thesis extends, and its published scope is worth stating precisely. C2S-Scale evaluates it on three benchmarks: cytokine stimulation of peripheral blood cells, bulk L1000 profiles, and the SciPlex3 small-molecule panel. In each the model receives an untreated profile together with a description of the treatment and produces the treated profile, so the setup is conditional rather than label-only generation. Two features of it are relevant to the comparison this thesis draws. Pairing is by group rather than by cell — for the bulk benchmark, treated and untreated samples are matched by cell line name — which is unavoidable given that sequencing destroys the cell it reads. And the single-cell results are scored as distributions: for each condition the model generates as many cells as were observed, and the generated population is compared against the observed one, which asks whether the model produces a plausible population for a condition rather than whether it distinguishes that condition from another.

The checkpoint fine-tuned in every experiment below is `C2S-Scale-Pythia-1b-pt`, the pretrained member of that family, whose training corpus is atlas data alone and contains no drug-treated cells.

2.3 The dispute: do these models beat uninformative baselines?

The field contains an unresolved and unusually sharp disagreement about whether any of this works, and this thesis is best read as an attempt to settle one corner of it.

On one side sits a series of benchmarks reporting that deep perturbation models do not beat trivial predictors. Ahlmann-Eltze, Huber and Anders [8] find that deep-learning prediction of genetic perturbation effects does not outperform simple linear baselines. Csendes et al. [9] reach the same conclusion for foundation cell models on post-perturbation RNA-seq, and Bendidi et al. [10] put it most bluntly in a title, reporting that a single PCA still rules them all. Kernfeld et al. [11] concur in a broad comparison of expression-forecasting methods. The pattern is not confined to transcriptomic readouts: on scalar drug sensitivity, Hauptmann and Kramer [12] hold the experimental protocol fixed across architectures and find that the measured differences between them largely dissolve, and DrEval [13] finds that deep models barely outperform a naive predictor built from mean drug and cell-line effects. The recurring result is that a per-perturbation mean is remarkably hard to beat.

Against this, Miller et al. [14] argue that these conclusions "largely stem from limitations of benchmarking metrics, not from the models themselves", and the argument is methodological rather than rhetorical. They formalise the use of positive and negative controls in perturbation benchmarking and introduce a new positive control, the *interpolated duplicate*, which blends a technical duplicate with the mean baseline using weights drawn from per-gene differential-expression significance — giving a reference point of known and tunable quality. They then define a quantitative measure of whether a metric can see that quality at all, the *Dynamic Range Fraction* (DRF), and apply it across 14 Perturb-seq datasets and 13 metrics. Widely used measures such as MSE and control-referenced Pearson correlation turn out to be poorly calibrated, while weighted and rank-based alternatives — weighted MSE, weighted $R^2\Delta$ [15], and Normalised Inverse Rank [16] — are consistently calibrated. Evaluated on the metrics that survive, they report, deep models *do* beat mean, control and linear baselines.

What is actually in dispute is narrower than it first appears, because both sides agree the standard metrics are defective. The disagreement is about what a corrected evaluation would reveal. One reading is that the models were always capturing real biology and the broken metrics were hiding it, so better measurement should expose genuine skill. The other is that the models were never capturing much, and better measurement will simply show that failure more clearly and more honestly. Put that way the question is empirical rather than interpretive — but it is settled only within the setting where it is tested, and the setting of this thesis has been tested by neither camp. Miller et al. and Ahlmann-Eltze et al. both study **genetic** perturbation, where an intervention is a designed, near-binary knockout with a named target gene. DrEval studies **drug sensitivity** as a scalar potency such as IC_{50}, predicted from cell-line omics. None of them studies a full transcriptomic response to a small molecule — which acts at a dose and typically engages several proteins at once — at single-cell resolution, produced by an autoregressive generative model.

The position taken here is to adopt the proponents' instrument rather than argue with it. Section 3.2 ports the DRF framework onto drug perturbation and uses it to decide which metrics are admissible in this setting before any model is scored, so that the evaluation is chosen by the sceptics' opponents rather than by us. Whatever the models then turn out to do, the measurement cannot be dismissed by the objection Miller et al. themselves raise.

2.4 Six ways a drug-response benchmark can mislead

DrEval [13] deserves separate treatment, because beyond its verdict it supplies a diagnostic vocabulary that this thesis adopts throughout. Auditing drug response prediction, it identifies six obstacles to clinical translation: non-reproducible models, data leakage producing weak generalisation, pseudoreplication, biased evaluation metrics, missing ablation studies, and inconsistent response data. Four of them shape the methodology of Section 3 directly.

Pseudoreplication, in Hurlbert's original sense [17], is treating repeated measurements of one unit as independent observations. DrEval observes that GDSC2's 395,025 measurements descend from only 969 cell lines and 287 drugs, so an interval computed over measurements is badly overconfident about how much independent evidence exists. Our data has the same shape — hundreds of thousands of cells resolving to a few thousand conditions across 50 cell lines — and every confidence interval in this thesis is consequently a clustered bootstrap over cell lines rather than over cells.

Biased evaluation they frame as a version of Simpson's paradox. Because different drugs act at vastly different concentrations, most of the variance in a response dataset lies *between* drugs, so a model that learns nothing but each drug's mean will explain most of it. Their illustration is stark: a global Pearson correlation of 0.90 alongside an average within-drug Pearson of 0.01. Section 3.2 explains why a discrimination metric evaluated *within* a comparison set is immune to this by construction.

Their treatment of splits is the part we borrow most directly. DrEval names four, each corresponding to a different application: Leave-Pairs-Out for missing-value imputation, Leave-Cell-Lines-Out for personalised medicine, Leave-Drugs-Out for drug design, and Leave-Tissue-Out for repurposing. Our holdout maps onto two of these — `unseen_combo` is Leave-Pairs-Out and `unseen_drug` is Leave-Drugs-Out — and Section 3.11 uses their names. The mapping carries a warning we take seriously. Citing Partin et al. [18], DrEval reports that good Leave-Pairs-Out and Leave-Cell-Lines-Out performance is reachable simply by predicting each drug's average response across the training cell lines, so a strong result on those splits demands that the per-drug average be measured explicitly rather than assumed away.

Their remedy for missing ablations is randomisation and robustness testing. The scramble ablation of Section 3.3 is such a test, specialised to the drug: hold every byte of the prompt fixed except the compound, and see whether anything downstream notices.

2.5 Connectivity mapping and signature transfer

The Connectivity Map programme [19, 20] built an entire drug-discovery methodology on a specific premise: that a compound induces a characteristic transcriptional signature stable enough across cellular contexts that querying a reference database by signature similarity is informative. The L1000 platform [20] operationalised this at scale, and its 978-gene landmark set defines the gene panel used here, intersected with Tahoe's measured genes to give 946.

The premise — that a drug's signature transfers across cell lines — is normally assumed rather than measured, and how much of it holds determines whether a conditional model has anything left to predict once a per-drug constant is accounted for. Section 3.9 measures it directly, corrected for measurement noise, at single-cell resolution.

2.6 Decodable, used, and the global workspace

Linear probing establishes that information is *present* in a representation, not that it is *used* by the computation, and the distinction is central to the analysis in Section 3.6.

Elazar et al. [21] make the methodological case for keeping the two apart: probe accuracy licenses no conclusion about behaviour, and the way to ask whether a property is used is to remove it from the representation and measure what the model then does. Their amnesic-probing design is the one adopted here, with the addition of a matched control, since an intervention that fails to move the output proves nothing unless a comparable intervention does.

A recent line of work gives that distinction a sharper form. Gurnee et al. [22] report that language models maintain a small privileged set of representations — which they term the *J-space*, identified by a Jacobian Lens that measures the average linearised effect of an activation on the model's likelihood of producing a particular token — and argue it functions as a global workspace analogous to conscious access. The J-space is strikingly small: roughly ten to twenty-five concepts are active at any moment, typically accounting for under 10% of activation variance. Its defining property is causal rather than correlational, and they establish it by intervention: swapping one active workspace vector for another changes the model's answer to match. Content in the J-space drives behaviour; content outside it may be richly represented and still inert with respect to what the model actually emits.

That framing matters here because it supplies both a hypothesis and an experimental design. The hypothesis is that a property can be decodable from the residual stream at high accuracy while sitting outside the workspace the generation readout consults, in which case no amount of encoding it more explicitly will change the output. The design is the swap itself: overwrite one candidate representation with another and ask whether the model's behaviour follows. Section 3.6 constructs a domain analogue of exactly this test, substituting a drug's learned representation for another drug's and measuring whether the emitted profile moves toward the substituted drug.

The related account of why a model might allocate its representations this way comes from Elhage et al. [23], who show that networks reserve dedicated directions for features the computation needs on every forward pass and push the remainder into superposition, where a feature remains linearly recoverable without occupying an axis of its own. Read together, the two suggest an asymmetry worth testing: a property required to emit any plausible output should sit in the causally active set, while one needed only for occasional, context-dependent composition need not.

2.7 Optimal transport for state matching

Predicting a treated cell from an untreated one runs into an obstacle that is not statistical but physical: sequencing destroys the cell it measures. No cell is ever observed both before and after treatment, so there is no ground-truth correspondence between the control population and the treated population — only two clouds of points in expression space, sampled from the same well at different times. Any construction of paired training data must therefore *infer* a correspondence, and optimal transport is the standard framework for inferring one.

The problem in its classical form asks for the cheapest way to move one distribution of mass onto another. Monge's original formulation seeks a map $T : \mathcal{X} \to \mathcal{Y}$ pushing a source measure μ onto a target ν, so that $T_{\#}\mu = \nu$, while minimising

$$\int_{\mathcal{X}} c\big(x, T(x)\big)\, \mathrm{d}\mu(x), \tag{2}$$

where c is a ground cost, typically squared Euclidean distance. This is intuitive — every source point is assigned one destination — but it is frequently ill-posed, because a deterministic map cannot split mass, and no such map exists when the target is more spread out than the source. Kantorovich's relaxation resolves this by replacing the map with a *coupling*: a joint distribution $\pi \in \Pi(\mu, \nu)$ whose marginals are μ and ν, giving

$$\min_{\pi \in \Pi(\mu,\nu)} \int_{\mathcal{X}\times\mathcal{Y}} c(x, y)\, \mathrm{d}\pi(x, y). \tag{3}$$

A coupling may send fractions of a source point to several destinations, which is what makes the relaxed problem always solvable and appropriate for noisy, overlapping empirical distributions. When c is the p-th power of a metric the optimal value defines the p-Wasserstein distance $W_p(\mu, \nu)$; the case $p = 2$ is the one used here, and its geodesics give a principled way to interpolate between distributions.

Solved exactly, the discrete problem is a linear program with roughly $O(n^3 \log n)$ cost, which is prohibitive at single-cell scale. Entropic regularisation makes it tractable. Adding a negative entropy term to the objective,

$$\min_{\pi \in \Pi(\mu,\nu)} \langle C, \pi\rangle \;-\; \varepsilon H(\pi), \tag{4}$$

renders the problem strictly convex, so the optimum is unique and takes the scaling form $\pi = \mathrm{diag}(u)\, K\, \mathrm{diag}(v)$ with Gibbs kernel $K_{ij} = \exp(-c(x_i, y_j)/\varepsilon)$. The Sinkhorn–Knopp algorithm recovers u and v by alternating matrix–vector products [24], which are cheap and parallelise well on a GPU.

The regularisation parameter ε is not merely a computational convenience, and it is the reason this framework appears in this thesis at all. It controls how sharply mass is assigned. As $\varepsilon \to 0$ the coupling concentrates and approaches the unregularised solution, in which each control cell is matched to essentially one treated cell. As ε grows the plan blurs, and in the limit every control cell is coupled equally to every treated cell — the product of the marginals. Given a coupling, the *barycentric projection* maps each control cell x_i to the π-weighted average of its partners,

$$\hat{y}(x_i) \;=\; \frac{\sum_j \pi_{ij}\, y_j}{\sum_j \pi_{ij}}, \tag{5}$$

and the two extremes of ε therefore produce two familiar objects: at small ε a specific observed treated cell, and at large ε the group mean. A sweep over ε is thus a continuous interpolation between two target constructions that would otherwise have to be treated as unrelated alternatives, which is how Section 3.7 uses it.

Applied to biology, this machinery has largely been used for trajectory inference. Waddington-OT [25] treats the cell population at each time point as a distribution in expression space and recovers a probabilistic ancestor–descendant link between snapshots, on the assumption that cells follow paths minimising transport cost. Subsequent work relaxes the assumption that mass is conserved — cells divide and die, so the two populations need not have equal total mass — using unbalanced formulations built on the Wasserstein–Fisher–Rao metric. A separate line of work exploits the fact that when both measures are Gaussian the transport cost has a closed form, the Bures–Wasserstein distance, which splits cleanly into a shift in means and a change in covariance structure and so distinguishes a population that moved from one that merely spread out.

Two practical points govern how couplings are computed here. The ground cost must be evaluated in a space where Euclidean distance is meaningful, and raw counts are not such a space: they are sparse, overdispersed and dominated by library size. Couplings are therefore computed in the released Tahoe scVI latent [26], a variational autoencoder embedding that models counts with a negative-binomial likelihood and yields a compact continuous representation in which distances behave. And because the cost of a full coupling grows quadratically, transport is solved within matched groups — a single cell line on a single plate — rather than across the atlas, which is both computationally necessary and biologically correct, since a control cell from one cell line is not a plausible counterfactual partner for a treated cell from another.

3 The investigation, in the order it happened

A model that has learned something about a drug should answer differently when it is told a different drug. That is the whole question, and it is cheap to ask — swap the compound named in the prompt, change nothing else, and see whether the prediction moves. Asking it honestly turned out to be expensive, because the first thing the investigation had to measure was its own instrument, and the instrument was reporting near-perfect scores for a predictor that had never been told which drug it was predicting.

What follows is the investigation in the order it happened. Every instrument here was built in response to the measurement before it, and that order is kept rather than tidied away: a question, the tool that can answer it, what the tool showed, and what that showing forced next. Design choices appear as the failures they avoid rather than as best practice — nearly every one was forced by a specific way the obvious alternative breaks on this data. Definitions appear where they are first needed. Dead ends stay in, including three substitution designs that failed and two printed claims that had to be withdrawn, because the confound that killed each one is the reason the surviving design looks the way it does. Every section closes with its central claim, set off from the text, in language that could be shown wrong. Two conventions hold throughout: every number is quoted with its interval where one exists, and "not established" and "shown to be zero" are different statements — the difference is load-bearing in several places.

The findings, stated before the apparatus that produced them. First, the field's standard perturbation metric is saturated by a predictor that carries no information at all, so the instrument was repaired before the model was measured. Second, on the repaired instrument the model is drug-blind: it returns the same answer whether told the true drug or a different one, and a baseline that copies the untreated cell matches it exactly. Third, the blindness is not an encoding failure. Drug identity is written into the model's activations, increasingly explicitly with depth, and the readout barely consults it. Fourth, re-encoding the training target as the drug-specific residual makes the model demonstrably sensitive to the drug in its prompt — yet that sensitivity does not become reliable prediction on held-out treatment wells, and a lookup table of training averages comes within two points of the replicate ceiling where the model covers under a fifth of the same distance. The argument is a dissociation between prompt sensitivity and biological fidelity. Figure 1 is the map.

3.1 The atlas, the model and the rules of measurement

The corpus is Tahoe-100M [1]: drug-treated single-cell transcriptomes across cancer cell lines, with plate-matched DMSO controls. Metadata supplies cell line (Cellosaurus ID), drug, dose (`drugname_drugconc`), plate, and a fine-grained mechanism-of-action annotation (`moa-fine`). All analysis is restricted to the $G = 946$ genes in the intersection of the

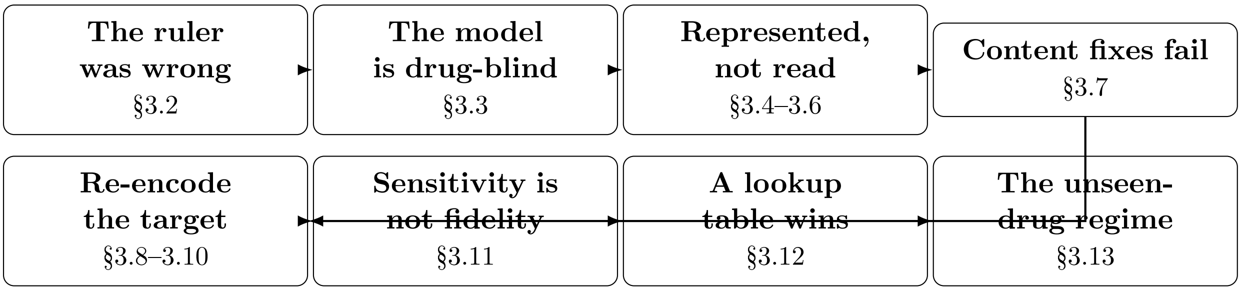


Figure 1: The investigation in eight beats. Each is told with the instrument it required.

L1000 landmark set [20] with Tahoe's measured genes, library-normalised to counts per 10,000 and transformed as $\log_{10}(1+x)$. The panel is a budget choice rather than a biological one: a cell is represented as a token sequence, the model's context is 8192 tokens, and a prompt must carry an instruction line and two cells, so the full transcriptome does not fit. The normalisation is what makes a rank representation meaningful. Without library normalisation a sequence's ordering tracks sequencing depth; without the log transform a handful of ultra-high genes occupies the head of every sequence regardless of condition.

One representational pitfall is worth recording because it is silent. Tahoe's per-cell `genes` field holds vocabulary token identifiers, not positional indices into the gene metadata table. Mapping them positionally scrambles gene identity quietly: the values exceed the metadata row count, so the error surfaces only as an out-of-bounds index at the tail. All expression construction here maps `gene_metadata['token_id']` to gene symbol to panel index, and the construction is validated by a PCA cell-line probe on the resulting panel, which reaches 0.864 against a chance rate of 0.02.

The physical layout of the atlas drives every inferential choice in this chapter, so it is stated before anything that depends on it. Tahoe applies one drug at one dose to one *well* containing roughly 49 cell lines simultaneously: 289 treatment samples in all, a mean of 48.6 cell lines per well (median 49, range 43–50), with 286 of 287 distinct (drug, dose) treatments occurring in exactly one well. Each drug is assayed on its own plate or plates, roughly 4–12 drugs per plate, with plate-matched controls. Three consequences run through everything below. Conditions from one well share a physical assignment, so they are not independent — this sets the inference rules below. Drug and plate are partially confounded, so discrimination must be measured within plate (§3.3). And a (drug, cell line) pair cannot be held out cleanly, because the pair's well remains in training through its other ≈ 48 lines (§3.11).

Three units of analysis run through everything that follows, and confusing them is the easiest way to misread a number. A *condition* is a (drug, cell line, plate, dose) tuple, the level at which responses are scored. The physical treatment assignment is the *treatment sample*, one treated well, so a condition is an observation nested inside an assignment rather than an assignment. Plates are numbered within cell line rather than globally, so the batch unit is the pair (cell line, plate) and never the plate label alone. A *pseudobulk* is the mean expression profile of a group of cells; a *control pseudobulk* is the same object over plate-matched DMSO cells. Dose is carried as a molar concentration resolved from `drugname_drugconc`, not as a unitless float: an earlier pipeline kept only the numeric part, which collides $1\,\mu$M with $1\,$nM. A direct audit of all 289 treatment samples found 289 usable molar doses with zero unit collisions: every drug in this atlas states its doses in a single unit, so the defect was a demonstrated possibility that did not materialise. Combination samples are dropped rather than analysed as their first component.

Table 1 states the quantities every n in this chapter should be located against. Two populations are used and they are not interchangeable. The broad characterisation of the task (§3.3) streams a wide sample of the atlas; the residual arm (§3.10 onward) works from a smaller cached subset built once and reused, so that every arm sees identical cells.

Task characterisation (streamed sample)	
conditions characterised	6,628
of which statistically active against DMSO	3,647 / 5,014 (72.7%)
identifiable in principle (own replicate ceiling ≥ 0.8)	3,064 (46.2%)
with a plate-mate closer than their own replicate	78.7%
median cells per condition	44
median #DEGs per condition (BH $q < 0.05$)	0
median SNR (effect $\div$ replicate noise)	0.75
drugs with a real dose series	218 / 351 (62.1%)
drugs seen in ≥ 3 cell lines	227
Residual-frame cache (rebuilt targets, §3.8)	
cells (treated / control)	620,564 (600,000 / 20,564)
(cell line, plate) groups	198
drugs / cell lines	106 / 50
conditions with ≥ 40 treated and ≥ 20 control cells	6,617
by split after inventory filters (train / held-out wells / held-out drugs)	4,999 / 900 / 711
retained by the data-resolved reliability line	2,523 of 4,999 train (50.5%)
training examples / well-held-out validation examples	147,710 / 3,600
Evaluation sets	
tier 1 (seen conditions) / tier 2 (unseen drugs) / tier 3 (unseen combinations)	4,325 / 2,188 / 99
conditions scored per tier in the generation evaluations	400
tier-2 drugs in the expression-frame NIR benchmark (within plate)	606
conditions scored in the residual-frame evaluation	570 (41 cell lines, 87 drugs)

Table 1: The populations behind every n quoted in this chapter. The 72.7% active figure is over the 5,014 conditions with enough cells for the permutation test, not over all 6,628. The identifiability bar is a per-condition threshold — a condition's own replicate ceiling must clear 0.8) and is distinct from the aggregate ceilings of Table 2.

Three evaluation tiers, defined by what the training set contained, are used for the broad characterisation. *Tier 1* holds conditions whose (drug, cell line) pairing was seen in training, so it measures fit rather than generalisation. *Tier 2* holds conditions whose drug never appeared, which is Leave-Drugs-Out in the vocabulary of Section 2.4. *Tier 3* holds pairings whose drug and cell line were each seen but never together, which is Leave-Pairs-Out. Tier 3 is small (99 conditions) and is reported only where it is adequately powered; the residual arm builds its own, larger holdout, and the unit that holdout is keyed at turns out to be one of the most consequential choices in the chapter (§3.11).

The model side is equally compact. A cell is written as a *cell sentence*: its expressed panel genes in descending order of expression, terminated by `[END_CELL]`. The prompt supplies cell line, drug, dose, mechanism, and the untreated control cell's own sentence; the response is the treated cell's sentence. Two tokens, `[END_CELL]` and — for the residual arm — `[DOWN]`, are registered as atomic special tokens; left unregistered, `[DOWN]` splits into sub-words and the up/down boundary ceases to be a clean symbol. The backbone is `C2S-Scale-Pythia-1b-pt` [3], a C2S-Scale checkpoint built on the Pythia-1b decoder [7], cold-started for every arm from the same base checkpoint with identical hyperparameters (1 epoch, learning rate 10^{-5}, gradient accumulation 16, maximum length 8192, bf16), so that in each experiment the training target or objective is the only variable. What that checkpoint has already seen bears on how the held-out splits should be read. It is pretrained on over 50 million human and mouse cells from 825 single-cell datasets in CELLxGENE and the Human Cell Atlas — an atlas corpus, carrying no chemical or genetic perturbation data — and its model card lists cell and tissue prediction, conditioned generation and gene-set conversion among its capabilities, but not perturbation prediction. The base model therefore reaches our fine-tuning having never seen a drug-treated cell. A drug held out here is unseen by the whole pipeline, not merely by its final stage.

Vocabulary. Six terms carry the argument and are collected here before first use. Two *measurement frames* run through the chapter and are never interchangeable. In the *expression frame* the prediction is a cell profile, scored against other drugs on the same plate. In the *residual frame* (§3.8) the prediction is a drug-specific signature, scored against other drugs in the same cell line. The *comparison set* J — the conditions a prediction is ranked against — is what fixes chance at 0.50 in both frames, and it is declared explicitly at every score below. The *generic* is the drug-agnostic mean response subtracted to form the residual: the stress and cell-cycle programme common to all compounds, and the most important object in the chapter. The *residual target* is what remains once the generic is removed. It is built *split-before-fit* (the holdout is fixed from metadata before any statistic is estimated) and *leave-one-drug-out* (no condition's generic contains its own drug). The *scope* of the generic, `cell_line` or `plate`, is the set of drugs the mean is taken over, and §3.8 shows the choice is consequential. *Control-copy* is the baseline that predicts the untreated cell unchanged; it carries zero drug information by construction. The residual arm's split has three parts — `train`, `unseen_combo` (an unseen *well* of a seen drug) and `unseen_drug` — and samples conditions under an explicit per-split *quota*, because a uniform draw reproduces the pool's proportions and starves the held-out splits (§3.11). Five training *arms* (model variants differing only in target or objective) are used; they are enumerated in Table 20 of the Appendix. Where a claim below is multiplicity corrected, the family is those five arms by up to seven depths.

Ceilings. Almost every negative result in this chapter is stated as a distance from a *ceiling*, and the word names one estimator, not nine: the score a real replicate of the condition achieves when substituted for the prediction. What varies is the frame, the comparison set and the cell budget, and Table 2 says which is which. Two properties matter for how the ceilings can be read. They are *split-half* estimates rather than biological replicates, because Tahoe places 286 of 287 treatments in exactly one well — a biological replicate does not exist in this atlas. And a split-half ceiling is a within-well sampling-precision estimate, which is optimistic relative to true biological reproducibility; the two coverage figures quoted against the residual-frame ceiling in §3.12 inherit that optimism.

Ceiling	Frame and metric	Comparison set and budget	Where used
0.76	expression, DE-Δr	two real cells	metric audit (§3.2)
0.625 / 0.575	expression, NIR	cross-plate / within plate	plate leak (§3.3)
0.67–0.83	expression, forced choice	within plate, by cell budget	§3.3
0.576	expression, NIR	within plate, tier 2	the blindness verdict
0.61 / 0.76 / 0.85	expression, NIR	within plate; 10/40/120 cells	identifiability curve
0.958	residual, NIR	within cell line, by split	§3.12
0.742 / 0.7432	token retrieval	within condition, split-half	§3.8

Table 2: One estimator, seven settings: every "ceiling" in this chapter is the score of a real replicate of the condition, and they differ only in frame, comparison set and cell budget. All are split-half estimates, because the atlas has no second well of any treatment to replicate against.

Rules of measurement. Two rules apply to every result below, plus one statement of provisionality. The first concerns uncertainty. Conditions are not independent, and they are not independent in two crossed ways at once: two conditions from the same treated well share a physical assignment and a batch, and two conditions from the same cell line share the biology. Treating conditions as independent gives intervals that are too narrow — here, about 1.3 to 1.5 times too narrow) and neither grouping alone captures the dependence. The tempting shortcut is actively misleading: clustering on the assignment unit alone uses the larger cluster count (289 wells against ≈ 50 lines) and would tighten every interval while sounding more rigorous. All intervals in this chapter are therefore two-way cluster-robust (Cameron–Gelbach–Miller),

$$V \;=\; V_{\text{line}} \;+\; V_{\text{well}} \;-\; V_{\text{intersection}}, \tag{6}$$

where the intersection of the two groupings is the single condition, so $V_{\text{intersection}}$ is the ordinary independent-sampling variance. When the estimator goes non-positive in a finite sample, the wider one-way variance is used and the branch recorded. A cluster-robust variance is asymptotic in the number of clusters, not observations, so the reference distribution is Student's t with degrees of freedom one less than the smaller cluster count. This matters where an arm spans only 25 wells, and it moves a borderline verdict in §3.11. For statistics computed over pairs of conditions, such as the transfer coefficient, neither grouping partitions the observations, a pair belongs to two nodes at once, so those intervals use the Fafchamps–Gubert dyadic estimator, under which two pairs are dependent whenever they share either member. Throughout, an estimate is quoted as value [lower, upper], a two-way cluster-robust 95% interval unless stated otherwise. This is the structure DrEval [13] identifies in this field more broadly, citing Hurlbert's pseudoreplication [17]: hundreds of thousands of cells resolving to a few hundred conditions over 50 cell lines.

The second rule concerns generation validity. Every generation-based evaluation reports the `[END_CELL]`/`[DOWN]` completion rate, the fraction of emitted symbols that are valid panel genes, and the duplicate rate, before any score is quoted. This is not hygiene: a 600-token generation cap silently truncated 26% of generations in an early run and halved the measured effect. Every contrast is additionally generated with a fixed per-(condition, arm) seed, so the two arms of a paired comparison are not drawn from different points of one global stream.

The provisionality is stated once. Every number derived from the model evaluation comes from the generation run identified in the project's run record; a corrected re-run is in flight, and while magnitudes will move, no direction is expected to change.

3.2 The metric is the first result

The investigation begins with the scoreboard, because every later claim depends on it. The field's default metric is a correlation between predicted and true change from control, restricted to the genes that actually move. Writing x for the control profile, y for the treated truth and $\hat{y}$ for the prediction, let

$$S_K \;=\; \arg\operatorname{top-}K_g\, |y_g - x_g| \tag{7}$$

be the K genes with the largest true change. Then the differential-expression delta-correlation is

$$\text{DE-}\Delta r \;=\; \operatorname{corr}\big((\hat{y} - x)\,\big|_{S_K},\; (y - x)\,\big|_{S_K} \big). \tag{8}$$

We report $K = 50$ with Pearson correlation throughout; $K \in \{20, 100, 200\}$ and the Spearman variant were computed and move the numbers by less than 0.02 except at $K = 200$, where the top-K set starts to include genes that do not move.

What does this metric reward? The measurement is a predictor that contains nothing: every gene placed at the middle rank, carrying no information about drug, cell or biology. It scores DE-$\Delta r(K{=}50)$ = 0.9999 [0.99991, 0.99992] — above the two-real-cell noise ceiling (0.76) and above the model (0.73). The ordering is inverted: less information yields a higher score, which is the signature of an artefact rather than of skill. The same predictor's partial-DE-Δr and panel-τ are undefined rather than merely poor: its shift from control is identically zero, so there is nothing left to correlate once the control is regressed out.

Predictor	Information	DE-Δr (K50, tier 2)	partial-DE-Δr (tier 1)
`revert_center` (genes $\to G/2$)	none	**0.9999**	undefined
`revert_mean` (predict mean control)	none, no fit	0.961	0.25
ridge linear (control $\to$ shift)	control	0.947	0.28
noise ceiling (two real cells)	—	0.76	—
model (1B LLM)	control + drug	0.73	not measured

Table 3: DE-Δr is saturated by a control artefact. Regressing the control out leaves $\approx$ 0.26 of real but drug-agnostic skill in the two baselines with a defined partial value, and a baseline that performs no fitting at all reaches the same figure. The partial column comes from the tier-1 run that computed it; the rest of the table is tier 2.

The cause is visible in the definition, and three properties of it, each innocent in isolation, compose into the failure. First, S_K is selected using the truth, so every predictor is scored on the genes the condition actually moved: the exam questions are chosen from the answer key. Second, treated cells in this atlas regress toward a shared response, so a prediction that reverts toward the population centre correlates with $y-x$ by construction. No drug knowledge is required to score well, only knowledge of where the centre is. Third, the fix is available in the same breath: partial-DE-Δr regresses the control profile out of both arguments before correlating, which removes the reversion route and is therefore the honest version of the same quantity. Removing the control collapses the raw 0.95 to $\approx$ 0.26 for the two baselines that were scored, and that residue is matched by a baseline that performs no fitting at all. The model's own partial-DE-Δr was never computed,[1] so the comparison is anchored instead to panel-τ, Kendall's τ between predicted and true rank over the full 946-gene panel, which selects no gene subset and is measured for every

[1]It requires a GPU pass that was not run, so the cell is marked rather than filled with the expected value.

arm. The anchor is used only to rank arms against one another on a fixed scale; no absolute claim rests on it, because the calibration audit below rules panel-τ inadmissible for absolute claims. It tells the same story: `revert_mean` 0.268, ridge 0.265, model 0.257 (tier 2, `worst` convention). The response decomposes into control reversion (an artefact), a generic drug programme (≈ 0.26, drug-agnostic, trivially matched), and a drug-specific component that no predictor captures.

A cell sentence lists only expressed genes, so a predicted profile leaves some panel genes unranked, and any rank-based metric needs a convention for the absent tail. Two were used throughout and reported side by side: `worst` assigns every absent gene the last rank G, and `francesca` assigns them a fixed middle rank $G/2$. The choice is immaterial — spike-in DE-Δr reads 0.952 against 0.954 under the two, and no baseline or ceiling in this chapter moves by more than 0.01 — so all figures are quoted under `worst` and the convention is not mentioned again.

The metric this investigation relies on instead is the Normalised Inverse Rank (NIR), introduced by Wu et al. in PerturBench [16] (there called the "transposed-rank" metric) and identified by Miller et al. [14] as one of the few consistently calibrated metrics across 14 Perturb-seq datasets. Figure 2 shows the construction. For a prediction $\hat{y}$ of condition i, with truths $\{y_j\}$ over the comparison set J,

$$\mathrm{NIR}(\hat{y}, i) \;=\; \frac{1}{|J|} \sum_{j \in J} \Big[\mathbf{1}\{\sigma(\hat{y}, y_i) > \sigma(\hat{y}, y_j)\} + \tfrac{1}{2}\, \mathbf{1}\{\sigma(\hat{y}, y_i) = \sigma(\hat{y}, y_j)\} \Big], \tag{9}$$

the fraction of the other conditions in J whose truth the prediction resembles less than its own, with ties counted at one half. The comparison set is declared at every score: conditions are grouped by (cell line, plate) in the expression frame and by cell line in the residual frame, and the two are not interchangeable: widening J across plates moves a drug-agnostic baseline from 0.161 to 0.399 (§3.3). Two properties do work that no other candidate does. Chance is 0.50 by construction, not empirically: it is a statement about J and nothing else, so "this model is at chance" is a statement rather than a comparison against a fitted null, and that is what makes the negative results in this chapter quotable without a control arm. And because the generic response is present in every condition's truth in equal measure, it cancels in the comparison: NIR is a discrimination metric, and only the drug-specific difference decides it, which is precisely the component the reversion artefact exploits. The tie term is not cosmetic. In residual space (§3.8) a predictor of "the average drug response" is exactly the zero vector, every similarity against it is zero, and a strict inequality scores it 0.000 rather than the 0.500 it deserves.

Our instruments compute NIR under two similarity functions and store both: `nir_expr`, Euclidean distance in expression space, and `nir_rank`, the same construction over rank vectors. Every NIR figure in this chapter is `nir_expr`, and the choice is load-bearing. Under `nir_rank` the tier-2 aggregate places every arm below chance, including the noise

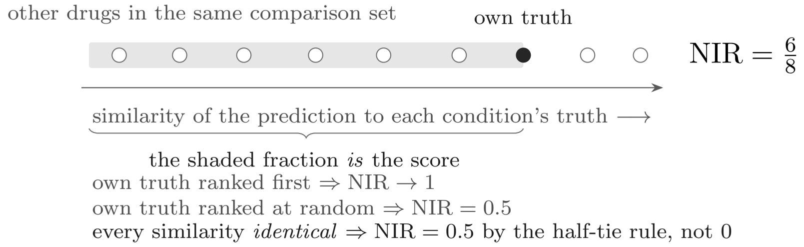


Figure 2: **NIR asks whether a prediction resembles its own condition's truth more than it resembles the truths of the other conditions in the comparison set J.** Own truth ranked first gives NIR → 1; ranked at random gives 0.5; and a predictor whose similarities are all identical — the zero vector, which is what "predict the average drug response" becomes in residual space — scores 0.5 by the half-tie rule, not 0.

ceiling itself (0.380, against the model's 0.313 and a drug-agnostic linear map's 0.285). A metric on which a real replicate of a condition fails to be retrieved above chance is not measuring what its name suggests. Nor does the choice contradict the calibration literature's preference for rank-based metrics: the rank in NIR's construction is a ranking of *conditions* by similarity, not of genes, and the similarity itself is computed in expression space. The DRF result below is computed on the expression-space variant; we report the calibrated one and name it here so the stored artifacts cannot be misread.

Choosing NIR over its rivals is itself a measurement, made with the framework of Miller et al. [14], which grades each metric rather than each model by asking how well it separates a positive from a negative control. The Dynamic Range Fraction is

$$\mathrm{DRF}(m) \;=\; \frac{m(\mathrm{pos}) - m(\mathrm{neg})}{m(\mathrm{perfect}) - m(\mathrm{neg})}, \tag{10}$$

on a scale where 0 is the drug-agnostic baseline and 1 a perfect predictor. Here pos is an interpolated-duplicate noise ceiling — a technical duplicate interpolated toward the mean baseline with per-gene weights given by differential-expression significance — and neg is a drug-agnostic leave-one-out mean. A negative DRF means the metric scores the uninformative baseline above the noise ceiling: it is inverted, and rewards the generic gene ordering rather than the drug. Three hardening choices matter for how the result can be read. Tied expression values receive average ranks, because breaking ties by array index makes a tied gene's rank depend on its position in the panel. The DRF ratio is recomputed whole inside every bootstrap resample, because its denominator is estimated from the same data as its numerator and holding it fixed understates the interval. And the family of five metrics is controlled simultaneously with Holm's procedure, because the claim "only NIR is calibrated" is a claim about all five at once.

Under the hardened protocol, NIR is the only calibrated metric. Its DRF is +0.635 [+0.618, +0.652] (Holm p = 0.0025; all-plates calibration run, 1,820 drugs over 25 cell lines). The four rivals remain strongly negative and none approaches significance (Holmadjusted $p = 1.0000$ for all four); their values come from the within-plate run (655 drugs over 61 cell-line×plate groups) and read -0.081 (`weighted_r2`), -0.231 (panel-τ), -0.415 (`spearman_expr`) and -0.448 (DE-Δr). The inversion is a property of the data, not a leak: a leave-one-out correction barely moves the numbers, since real cell lines carry 20–204 drugs, and it survives holding the plate constant.

This replicates Miller et al. — metric choice dominates, the rank-based metric survives, and the correlation-based metrics they flag are exactly the ones that fail here — and then diverges from them. They conclude that under calibrated metrics deep models outperform uninformative baselines; §3.3 evaluates on the calibrated metric and finds the model at chance. Two differences in setting are candidates for the discrepancy — genetic versus drug perturbation, and a discriminative regression model versus an autoregressive generative one — and §3.12 supplies a third and more structural possibility. Adopting their framework is also deliberate in a second sense: evaluating our model on the metric their analysis endorses means the drug-blindness result cannot be dismissed on the grounds their own paper raises.

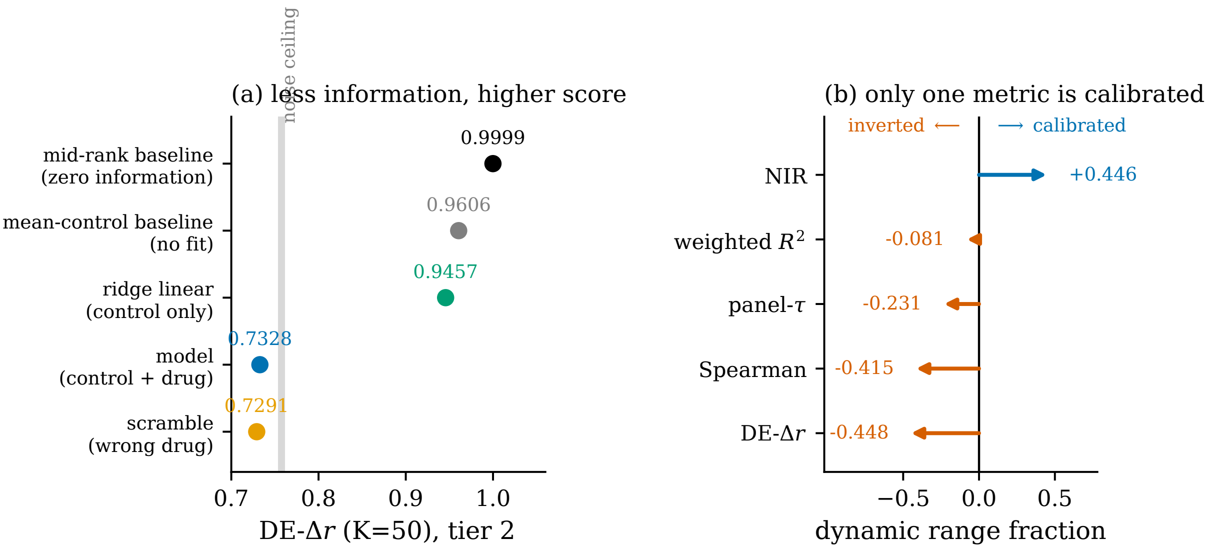


Figure 3: **Under the standard metric, less information yields a higher score.** (a) A predictor carrying nothing about drug, cell or biology scores DE-$\Delta r = 0.9999$, above the two-real-cell noise ceiling and above the model. Tier 2, $n = 400$ conditions. (b) Grading the metrics rather than the models: only NIR points forward. Hardened estimate +0.635 [+0.618, +0.652], all-plates run (1,820 drugs, 25 cell lines), and +0.519 [+0.479, +0.557] when partners are restricted to the same plate — the confound-free version, lower but still the only calibrated metric of the five; the four rivals shown are the within-plate run (655 drugs, 61 groups). `weighted_r2` carries no directional claim: its sign tracks cell count.

The metric can discriminate even if the model cannot, and the instrument that shows it is a spike-in forced choice: within one plate, a reference pseudobulk drawn from drug A's cells must be scored more similar to a disjoint sample of A's own cells than to a sample of drug B's, over many resamples, with chance at 0.50; contaminating B's population with a growing fraction of A's cells titrates the difficulty, so the curve of accuracy against contamination characterises each metric's sensitivity. At pseudobulk and zero contamination the separation is essentially perfect (panel-τ = 1.000, DE-Δr = 0.995, top-n τ = 0.987, plate held constant). The instrument is not the bottleneck.

> THE CLAIM. The field's standard score is saturated by a predictor carrying no information (DE-Δr = 0.9999); of the candidates audited, only NIR is calibrated (DRF =+0.635 [+0.618,+0.652]); and on NIR the instrument can discriminate. Every drug-use claim in the rest of this chapter is therefore made on the calibrated metric.

3.3 The model is drug-blind

With a ruler that cannot be gamed in hand, the investigation can ask the question it exists to answer: does the model use the drug at all? The answer is no, and it is stated before the instruments because the instruments matter only as the objections they close. On the calibrated metric, within plate on tier 2 ($n = 606$), the model scores 0.498 against a replicate ceiling of 0.576. It is matched by predictors that know nothing about the drug: a drug-agnostic linear map scores 0.500, a control-copy 0.504, the global mean 0.180. And on the identifiable stratum, where the task is provably winnable, a zero-drug-information control-copy reaches 0.766 against the model's 0.768.

Four instruments produce this verdict, and each exists because a specific objection would otherwise be fatal. They are the benchmark itself (the model against drug-agnostic baselines on the calibrated metric, within plate); the scramble ablation (the only manipulation that isolates drug use); forced-choice grading (a truth-based check whose chance value is fixed); and output invariance (a truth-free check that no scoring choice can confound). The benchmark is the verdict above; the other three are built and reported below.

One measurement rule comes first, because the verdict depends on it. Drug and plate are partially confounded by the atlas's design (§3.1), so discrimination grouped by cell line alone lets the batch signature stand in for drug identity. Measured directly, by scoring the same tier-2 conditions with and without the plate held constant:

The mean baseline is the clearest demonstration: it contains no drug information whatsoever, and it scores 0.399 when the comparison set may span plates against 0.161 when it may not. Every discrimination instrument therefore gained a `--same_plate_only` mode and all discrimination results in this chapter were re-run within plate. Conclusions are unchanged; some magnitudes shrink.

The central manipulation is the *scramble ablation*: the prompt is left byte-identical

Predictor (drug-agnostic)	cross-plate	within plate
mean baseline	0.399	**0.161**
control-copy	0.551	0.510
noise ceiling	0.625	0.575
fraction of drugs at chance	43%	49%

Table 4: Expression-frame NIR. A drug-agnostic mean baseline more than doubles its score when the plate is free to vary, which is what plate leakage looks like. The ceiling falls once the plate is held constant because the batch signature was inflating it too.

except that the drug name and mechanism are replaced by another drug's, while the control cell and the scoring truth are unchanged. Because both arms share the same control cell, control- and batch-derived leakage cancels in the scramble gap $\Delta_{\text{scr}} = \text{NIR}_{\text{model}} - \text{NIR}_{\text{scramble}}$. This is the only manipulation that isolates drug use, and every drug-use claim in this chapter is anchored to it rather than to a comparison against a drug-agnostic predictor.

Two constraints on the swap partner are required for a gap measured this way to mean what it says, and both exist because the unconstrained pool fails in a specific way. The partner must be a different drug: allowing the same drug at another dose or plate substitutes the drug for itself, an unchanged output is then correct rather than blind, and the contamination concentrates in the most similar stratum because a drug's own other doses are usually its nearest neighbours. And the partner must be on the same plate: residuals retain plate structure, so a cross-plate partner differs from the target in batch as well as in drug, and the gap absorbs both. A scramble that swaps in a drug with a similar true response is also a weak test, and this is a real risk here because same-mechanism drugs are barely more similar than random pairs (within-MoA distance 2.322 versus between-MoA 2.377, ratio 0.977), so "different mechanism" does not imply "different response". Two remedies are used. A swap-distance sweep — two hundred (drug, swap) pairs over twenty cell lines on tier 2, binned by quantile of the response distance $\|y_A - y_B\|$ between the real and swapped-in drug — turns the single test into a curve of Δ_{scr} against that distance. And a stratified scramble defines three partners per condition, chosen by residual cosine within the cell line: `near` (most similar), `orth` ($\cos \approx 0$) and `opposite` (most anti-correlated). The strata make visible an assumption that a single comparator hides. The gap has two parts — how much naming the truth helps, and how much naming a lie hurts) and only the first is wanted. A usable null is therefore a stratum whose own score sits at chance. §3.11 shows this is true of exactly one of the three, and not the one an evaluation would naturally default to.

The third instrument is forced-choice grading: a prediction is scored against its own condition's truth and against another drug's truth from the same comparison set, recording how often the own truth wins. Chance is 0.50 by construction, and the ceiling — the

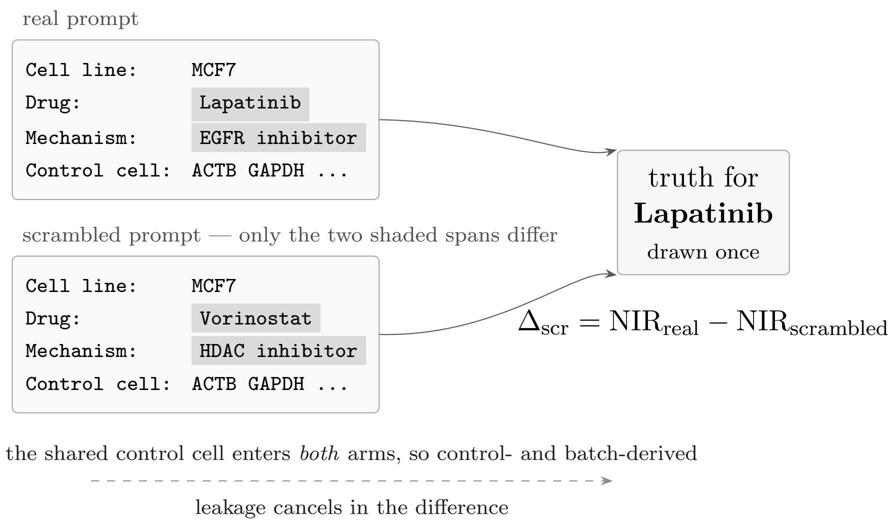


Figure 4: **The scramble replaces the drug name and its mechanism and nothing else.** Both arms are conditioned on the same control cell and scored against the same truth, so any leakage through the control cell or the plate enters both arms in equal measure and cancels in the difference Δ_{scr}. This is why Δ_{scr}, rather than a comparison against a drug-agnostic predictor, is the quantity every drug-use claim in this thesis is anchored to: the drug-agnostic comparison is not leak-immune, and a drug-agnostic mean baseline scores NIR $= 0.399$ when the comparison set may span plates against 0.161 when it may not, precisely because it inherits that leakage. A refinement matters here and is developed in Section 3.3: if the swapped-in drug happens to have a *similar* true response then an unchanged output is correct rather than blind, so the swap partner is chosen from three defined similarity strata and the evidence is the *gradient* across them rather than any single value.

same forced choice with a real held-out replicate substituted for the prediction — lands between 0.67 and 0.83 depending on cells per condition. The fourth is output invariance, which asks whether the drug token changes the generated text at all, without reference to any truth. For a fixed control cell, the model generates twice from the same prompt and once from each of several prompts differing only in the drug, and the instrument compares

$$\text{invariance gap} \;=\; \underbrace{\overline{\text{sim}}(\text{same prompt, resampled})}_{\text{sampling noise alone}} \;-\; \underbrace{\overline{\text{sim}}(\text{prompts differing in the drug})}_{\text{noise plus any drug effect}}, \tag{11}$$

with similarity measured as top-100 Kendall τ and as Jaccard overlap. A positive value means two generations for the same drug resemble each other more than generations for different drugs do; zero means swapping the drug perturbs the output no more than resampling at the same temperature does. Reported over 8 contexts $\times$ 12 drugs at temperature 0.8. This instrument is the weakest of the four — it can only detect whether the output moves, not whether it moves correctly — but it is also the least able to be confounded, since it needs no ground truth at all.

Instrument	Quantity	Value	Reading
Forced-choice grading	own-truth win rate	0.48 ($\approx$ scramble; ceiling 0.67–0.83)	chance
Output invariance	invariance gap	0.000 [-0.019, $+0.016$]	null
Scramble, tier 1	DE-Δr	0.740 real vs 0.739 scrambled	identical
Scramble, tier 2	DE-Δr	0.733 real vs 0.729 scrambled	identical
Scramble (identifiable subset)	Δ_{scr} (NIR)	$+0.014$ [-0.016, $+0.042$]	null

Table 5: Four independent instruments agree. Near-ceiling DE-Δr is achieved identically with a wrong-mechanism drug, so the score is drug-independent.

Four objections to the verdict close on measurement rather than argument. First, *is there anything to know?* A ladder of mean-shift baselines — each predicting the average response of a progressively narrower group — answers that directly. On tier 2, predicting the global mean scores DE-Δr 0.056; conditioning on cell line raises it about two-and-a-half-fold to 0.142; conditioning on mechanism does not move it (0.052); and conditioning on mechanism and cell line is no better than cell line alone (0.120 against 0.142). Tiers 1 and 3 tell the same story. The informative conditioning variable in this data is therefore the cellular context, not the compound class. It is worth holding onto, because §3.6 finds precisely that asymmetry inside the model's activations, and §3.13 finds that mechanism does carry signal once the generic programme is subtracted. The apparent tension is a frame difference rather than a contradiction: in full-profile space the generic response dominates and swamps a mechanism effect that becomes visible only in the residual frame.

Second, *is the task winnable at all?* Only 46.2% of (drug, cell line) conditions are identifiable in principle; 27.3% of tested conditions are statistically inert against DMSO and 78.7% have a plate-mate closer than their own replicate. But restricting to the identifiable subset does not rescue the model: it appears to beat the linear baseline there (0.768 vs 0.531), yet a zero-drug-information control-copy matches it exactly (0.766) and the leak-immune Δ_{scr} remains null. The model is drug-blind even where the task is provably winnable. The ceiling itself rises with aggregation (0.61 at 10 cells, 0.76 at 40, 0.85 at 120), so the signal is real and noise-limited.

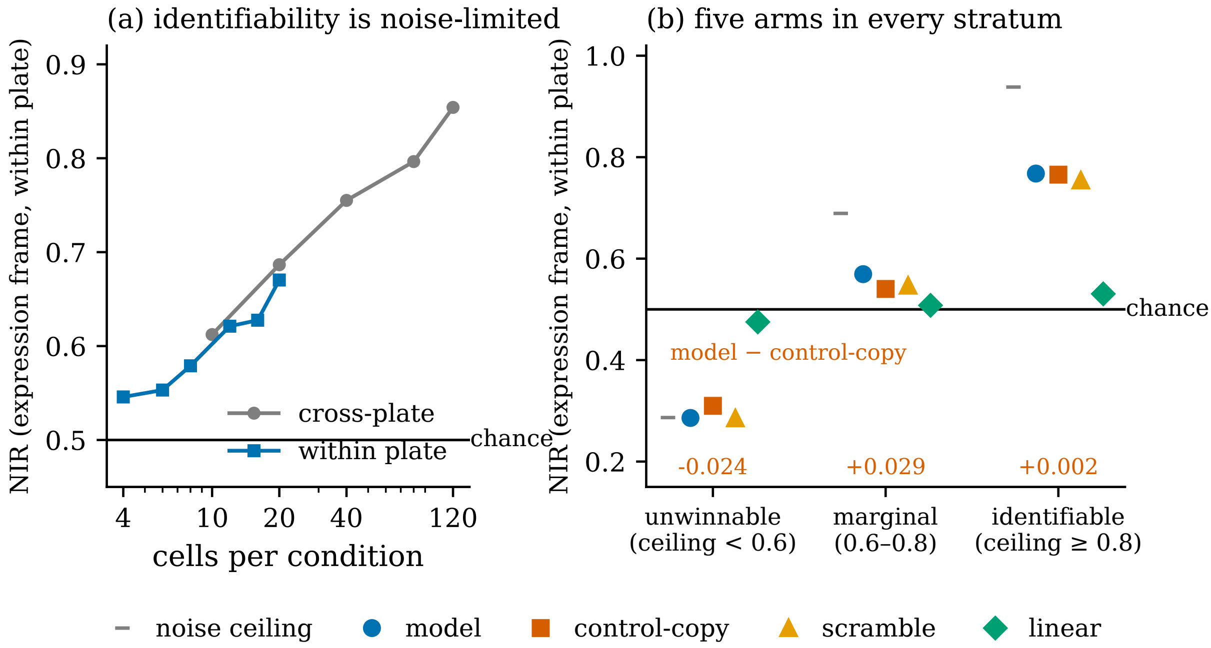


Figure 5: **The task is winnable where the noise ceiling is high, and precisely there, a predictor containing no drug information matches the model.** (a) The ceiling rises steadily with the number of cells per condition, cross-plate and within plate, so identifiability is noise-limited rather than absent: the drug signal is real and emerges under aggregation. (b) Conditions binned by their own ceiling, with five arms in every stratum. On the identifiable stratum ($n = 204$) the model reaches 0.768 — and a control-copy baseline that simply reproduces the untreated cell, and therefore knows nothing whatever about the drug, reaches 0.766. The paired contrast beneath each stratum is +0.002 [−0.026, +0.028] there, clustered over cell lines. Showing the model alone across these strata would make the visual claim that it does better on winnable drugs; the control arm is what shows that it does not.

Third, *does the representation throw the drug away?* Cell sentences keep only the rank order of genes and discard expression magnitude, and if the drug lived in the magnitudes the diagnosis of the following sections would be attacking the wrong thing. The objection is testable without any model, by asking whether true expression discriminates drugs any better than rank does. It does not: at single-cell resolution, cosine, Pearson and Spearman similarities on true normalised expression all sit within 0.04 of chance (0.529–0.531 [0.515, 0.543]), the same near-chance regime as rank. What recovers the signal is aggregation, not representation: at fifteen cells the control-referenced shift cosine reaches 0.793 [0.775, 0.809]. The drug is not hiding in the magnitudes, which matters twice over. It rules out the simplest alternative explanation for everything that follows, and it means the repair of §3.10 cannot be read as "restoring the magnitudes": that repair keeps rank throughout and changes only what is being ranked. The same measurement bounds the task from below — at single-cell resolution the same-drug versus different-drug discrimination gap is only $+0.006$ $[+0.003, +0.008]$, so whatever a model can learn here, it cannot be learned from a single cell.

Fourth, *does the scramble swap in similar drugs?* The swap-distance sweep resolves this: Δ_{scr} is null in every distance bin including the far tail (-0.019), across a $2.3\times$ spread in swap distance, with a trend correlation of $+0.051$. Telling the model it is a drug whose true response is maximally different still does not change the output.

> The claim. On the calibrated metric the model is drug-blind, and even where the task is provably winnable: at chance overall (0.498 against a 0.576 replicate ceiling), matched exactly by a zero-drug-information control-copy on the identifiable stratum (0.768 against 0.766), and unmoved even when told the most dissimilar drug on offer.

3.4 Is the drug represented at all?

A null result invites two readings, and they call for opposite repairs. Either the drug never enters the model's computation — an encoding failure, which target- and input-side fixes might reach, or it enters and is ignored, a readout failure, which only an objective-side fix could reach. Every behavioural measurement so far is consistent with both, because both predict the same output. The previous section has already closed a third possibility, that the task itself cannot be done: the ceiling rises with aggregation, a third of conditions are identifiable in principle, and the model is at chance even on those. So the failure is the model's, and the useful question becomes where in the model it lives. That question cannot be answered behaviourally; it needs the weights. It decomposes into three sub-questions, each with its own instrument, because each instrument's blind spot is exactly what the next must close. Is drug identity present in the activations? Does the network use the region where it is written? And when it uses that region, does it use which drug is written there, or only that something is? This section answers the first; the next two take the others in turn.

The first instrument is a linear decode probe, and its logic is deliberately weak: if even a simple linear reader can recover the drug from the activations, then the information is present in a form the network could in principle use. No claim is made yet about whether it does. The construction uses forward passes only, no generation and no training of the model itself. Real tier-2 evaluation prompts are grouped by drug; for each, the residual-stream activation is captured at the last prompt position, the position from which response generation begins, at every layer. A cross-validated multinomial logistic-regression classifier then learns to name which of the twelve drugs the prompt named, and its accuracy is compared against the floor measured on shuffled labels ($1/12 \approx 8\%$). The twelve drugs are a random draw from the tier-2 drugs with at least forty scored cells each; the count sets the 8% floor and, in §3.6, bounds the drug subspace at rank 11. One limitation is stated rather than repaired: the cross-validation folds are stratified over prompts and not grouped by well, so well-mates can straddle a fold boundary. The accuracy is therefore read as a presence test, not a generalisation estimate, and the argument below rests on the dissociation between decodability and causal share rather than on the absolute figure.

Drug identity reaches the representation and survives in depth: the probe reads 82% at layer 9 and 76% at layer 16 against the $\approx 8\%$ floor. The between-drug to within-drug variance ratio collapses about three-fold across the final layers — the drug is present throughout, but geometrically de-emphasised as the output approaches. Decodability alone, however, cannot carry the argument: a probe is an external reader with access to information the downstream layers may never consult. Whether the network itself reads this region is a causal question, and it is the next section's.

> THE CLAIM. Drug identity is linearly readable from the residual stream deep into the network — 82% against an 8% chance floor at the peak layer — so the behavioural null of §3.3 is not an encoding failure. The drug is in the model; whether anything inside the model reads it is the next question.

3.5 Is the representation used?

Asking whether the network uses the drug region means intervening on it, and intervening requires localising it first. The second instrument is built in four steps, and each step exists because the unguarded version of the same measurement misled us once already.

Step one: where the drug lives. On 480 tier-2 prompts — twelve drugs, forty real cells each) at seven layers, the twelve per-drug mean activation vectors are centred on the global mean and stacked, and their singular value decomposition contributes its top ten orthonormal directions: V_{drug}, the slab of the residual stream in which drug identity is written. Twelve centred class means span at most eleven directions, so the slab keeps essentially the whole span, noisiest directions included; for a removal claim that choice is conservative. The layer grid is even (2, 4, 6, 8, 12, 16) plus layer 9, added after the raw probe peaked there. A cell-line subspace is built identically and serves throughout as the positive control, because the model demonstrably does use cell line (§3.3's grouping ladder).

Step two: purification. The raw drug slab is partly a batch slab. Drugs are confounded with plate and dose by the atlas's design, and probing dose from inside the drug subspace recovers it at 0.90–0.95 at every layer: read naively, the "drug" axis would partly be measuring the batch. A context subspace is therefore built from the union of cell line, plate and dose directions, and the drug axis is orthogonalised against all of it, giving V_{pure} — the part of the drug axis not explained by batch. Every causal claim below is made on V_{pure}, never on the raw slab.

Step three: the removal gate. Before any "ablating the drug changes nothing" result can be believed, the instrument must prove that the slab actually contains the drug. The decode probe's accuracy is measured before and after projecting the drug subspace out, and the fraction of above-chance signal removed must exceed 0.8 — otherwise the null is unfalsifiable, a measurement of nothing. The gate passes at every layer: accuracy falls from 0.41–0.82 to a constant 0.121 floor, removing 88–95% of the above-chance signal. The nulls below are real properties of the model, not broken hooks.

Step four: the causal measurement, in logit space. The true response is teacher-forced and its per-token log-probabilities are read off clean; a forward hook then projects the slab out of the residual stream at every position ($h \leftarrow h - (hV)V^{\top}$) and the log-probabilities are read again; the measure is the mean KL divergence from clean to ablated over the response tokens, on 60 prompts. Scoring in logit space rather than by generating is a deliberate repair of the project's first causal attempt, which injected a steering vector and generated: the steering vector was small against the residual-stream norm, its effect on the output sat under the sampling-noise floor, and its "no effect" was uninterpretable. The ablation is run for V_{pure} (headline), for a random subspace of matched dimension (the null), and for the cell-line slab (the positive control). The prompt count is small and no interval is quoted: the ablation is a sign-and-magnitude instrument, and the inferential weight of the probe arc is carried by the identity test of §3.6, which is intervalled throughout.

Layer	Decode (raw)	Decode (context removed)	Ablate pure drug (KL)	Random matched dim	Ratio
2	0.408	0.354	0.155	0.012	12.7×
4	0.585	0.429	1.563	0.084	18.6×
6	0.692	0.529	1.435	0.251	5.7×
8	0.779	0.569	0.596	0.042	14.1×
9	**0.821**	0.602	0.418	0.100	4.2×
12	0.754	0.873	0.133	0.035	3.8×
16	0.758	**0.883**	0.015	0.004	3.6×

Table 6: All seven layers pass the removal gate (88–95% of above-chance drug signal removed); chance is 0.083. The raw probe peaks at layer 9 and declines; with context removed, decodability rises monotonically to layer 16.

What the ablation shows is a dissociation, and Table 6 quantifies it. With cell line,

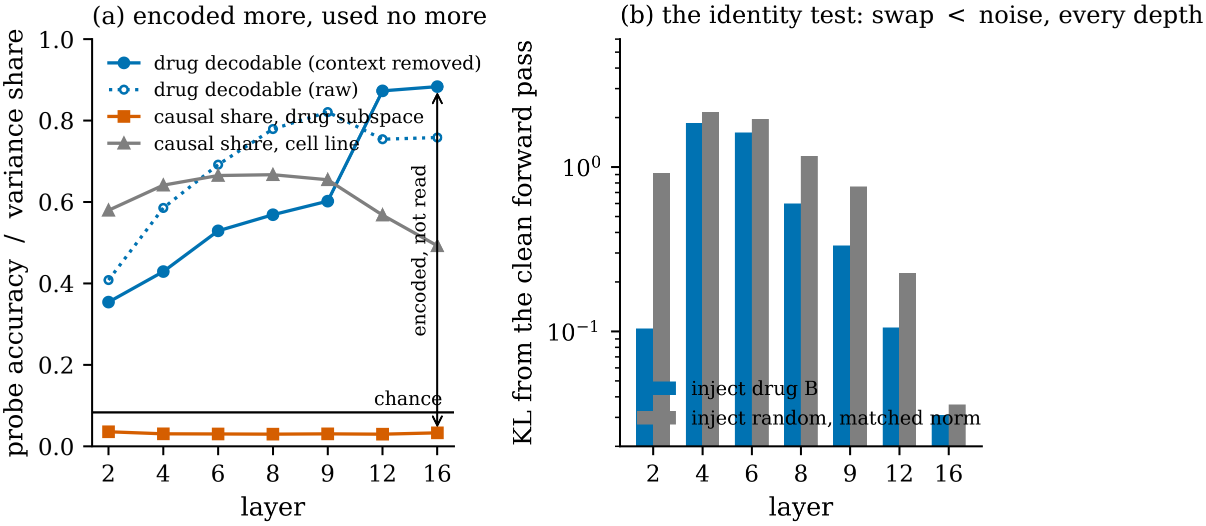


Figure 6: **The model computes an increasingly explicit drug representation and sets it aside.** (a) Context-removed decodability by layer, with the raw probe drawn for contrast. (b) Ablation of the purified drug subspace against a matched-dimension random subspace, on a log axis because late-layer divergences are small for everything. This is a functional-variance comparison; whether drug *identity* is read requires the substitution test reported in the text.

plate and dose projected out, drug decodability rises with depth, from 0.354 at layer 2 to 0.883 at layer 16: the representation becomes more explicit, not less. Yet the purified drug subspace's variance share stays flat at ≈ 0.03, against ≈ 0.6 for cell line, and ablating it costs 3.6–18.6× the matched-dimension random null across layers. That effect is real. It is also small beside the cell-line positive control, which the same ablation reads at 1.0–11.2× the drug's: removing the drug subspace costs 9–37% of what removing cell line costs at layers 2–9, rising to 98% at layer 12 and 25% at layer 16. In variance-share terms the gap is wider still, cell line carrying 15–22× the purified drug slab. The instrument is not the limit; the drug is a bit player.

This is also where ablation reaches its limit as an instrument. It proves the slab carries functional variance that downstream layers consume; it cannot prove that the slab's drug-specific content is what is read: the purified directions could carry generic strong-programme energy that merely co-varies with drug across twelve compounds. Whether the network reads which drug the slab encodes requires replacing what the slab says rather than removing it, and that substitution test is the next section's.

> The claim. The drug representation is used, but at a bit player's weight: decodability rises monotonically with depth while the slab's variance share stays flat at ≈ 0.03, roughly 5% of the cell line's ≈ 0.6, and ablating it costs 9–37% of what ablating cell line costs at layers 2–9. Ablation cannot say whether the slab's content is read — that needs a substitution.

3.6 Is the drug's identity read?

The substitution test is the only instrument that can answer this, and the design that survived audit is stated first; the three designs that failed are then the constraints it satisfies, one sentence each, with their algebra deferred to Appendix A.5. The test makes two structural choices so that its arms differ in nothing but which drug they name, and it ablates nothing. First, a contrast score,

$$\log P(g_B \mid \text{prompt}_A) \;-\; \log P(g_A \mid \text{prompt}_A)\,, \tag{12}$$

where g_A and g_B are real response strings emitted under drugs A and B by held-out cells. Any shift common to both responses, the entire logsumexp normaliser included, cancels algebraically. What survives is the model's preference between drug B's own held-out response and drug A's. Second, every arm is anchored at drug A as an on-manifold class-mean displacement of identical norm. Writing Π for the orthogonal projector onto the purified slab, the `swap` arm injects $\Pi(\mu_B - \mu_A)$, pointing at the drug whose response is scored; the `permuted` arm injects $\Pi(\mu_C - \mu_A)$, rescaled to identical norm — the headline control: same anchor, same construction, wrong drug); and an isotropic random direction in the slab is kept only as an off-manifold damage diagnostic, never as the comparison, because a random direction disrupts the model in ways an on-manifold class-mean displacement does not. The subtraction $\mu_B - \mu_A$ cancels the global mean exactly, so the injected vector is pure drug difference, and no ablation anywhere means there is no removed term for the control to trip over.

The three failed designs taught the constraints. A control drawn in the full 2048-dimensional stream puts $\approx 99.5\%$ of its energy where the intervention never acts, so the control must live in the same subspace as the treatment. Ablate-then-add delivers (added − removed), so an uncentred class mean turns the substitution arm into a restoration — hence no ablation, and an exactly centred displacement. A norm-matched random direction is off the class-mean manifold and inflates the logsumexp normaliser (measured +2.75 nats), depressing every target whether or not it carries identity; hence the contrast score, in which the normaliser cancels.

Everything is out of sample, and everything is on a comparable scale. The subspaces and class means are fit on one slice of cells; g_B is drawn from a third disjoint slice — never from the rows defining μ_B, else the injected vector and the scored response would share cells and the test would be partly circular; all scoring happens on the held-out slice. Because the five training arms emit different target formats, raw log-probability differences live on different scales, so every effect is divided by that model's own clean preference gap between A's and B's responses, measured with no intervention: the reported unit is the fraction of the existing preference gap the injection closes, comparable across arms.

Four guards carry the inference; the remainder — a drug-proxy screen, an injected-norm diagnostic and a three-world planted selftest — are engineering and are deferred to Appendix A.5. Uncertainty is a cluster bootstrap over ordered (A,B) pairs, because rows sharing a pair share both the injected vector and the scored response. Equivalence is by TOST: the interval must lie inside a pre-registered margin and contain zero, with the margin set at 10% of the model's own clean preference gap. A per-layer headroom gate excludes any layer where the cell-line positive control shows less than 5× dynamic range, because a saturated instrument can support neither verdict. And a magnitude ladder repeats the test at $\alpha \in \{0.5, 1, 2, 5, 10\}$ times the natural displacement, because a real identity effect should show a monotone dose–response.

This design was run across the five training arms of Table 20 and, for the two residual arms, replicated at three further seeds. Of 26 headline tests — five arms by up to seven depths, exactly one survives Benjamini–Hochberg correction: the residual arm at layer 12 ($q = 0.026$), with a monotone dose–response across the ladder's first three rungs (+0.0013, +0.0017, +0.0044 at $\alpha = 0.5, 1, 2$). The consensus arm at layer 2 ($p = 0.012$) does not survive ($q = 0.156$).

Arm / seed	effect	95% CI	p
residual, seed 42	+0.0197	[+0.0063, +0.0335]	0.0010
residual, seed 43	+0.0118	[+0.0060, +0.0173]	0.0003
residual, seed 45	+0.0085	[+0.0029, +0.0143]	0.0015
single-cell arm, six depths	nothing detectable at any depth (median −0.0012)		

Table 7: The identity test, as a fraction of that model's own clean preference gap between the two responses being compared (mean effect +0.0133 across the three seeds that measured it). Seed 44 produced no layer-12 measurement, it lost five of seven layers to the headroom gate, so the pre-registered criterion of two replications in three new seeds is met with the third a non-measurement rather than a failure.

The value of this is that it is a positive control on real data. The residual arm is independently known to use the drug: it is the arm whose prompt sensitivity is established behaviourally in §3.11. The probe detects identity reading there, replicably, and detects none anywhere in the arm that is behaviourally drug-blind. A null from an instrument that has never been shown to register a positive is uninterpretable; this one has.

> The claim. The generation readout consults the drug representation at very low gain: about 1% of the preference gap, an order of magnitude below the pre-registered margin for calling an effect material, in the one arm where it is detectable at all. The claim is not that the readout cannot read drug identity, and it is not the stronger mechanistic story that earlier drafts of this thesis told.

Three limits belong with it. The effect appears at one depth of seven, and layer 4 is positive in a single seed and null in the other three, which is what multiplicity discipline is for. The single-cell arm has no layer-12 measurement at all — its cell-line positive control reached 3.88× there, under the 5× headroom gate, so the layer was excluded — meaning the cleanest comparison, the same depth across two arms, does not exist; the dissociation rests on that arm being null at all six depths it could measure. And the sibling residual-holdout arm does not reproduce the effect (+0.0028 and +0.0049 in the two seeds where it was measured). Read together with the decodability series, the picture is a representation that becomes more explicit with depth while the process that writes the output barely consults it. That is suggestive of the workspace account of Section 2.6, and it is offered as an interpretation of the pattern rather than as a demonstration of the mechanism.

3.7 No fix that keeps the encoding works

The mechanism makes a falsifiable prediction, and this section is its test. If the defect is a readout that feels the magnitude of activity in the drug region but not its direction, then improving what sits in that region cannot help — and improving the training target is exactly that kind of intervention. The obvious objection is that one failed target construction proves nothing, so rather than try another we swept the family. Two constructions anchor the sweep. *Consensus* groups cells by (drug, cell line, dose) and averages them into one pseudobulk sentence, with every original prompt retained so only the response changes. *Optimal transport* computes, for each (cell line, plate) group, a coupling π between control and treated cells minimising $\sum_{ij} \pi_{ij} \|x_i - y_j\|^2$ under uniform marginals, solved by Sinkhorn iteration [24] on the kernel $\Gamma = \exp(-C/\varepsilon)$ with C the pairwise squared-distance matrix; couplings are computed in the released Tahoe scVI latent [26] ($n_{\text{latent}} = 10$), joined per cell by barcode rather than re-encoded — a re-encode produced a degraded latent (cell-line probe 0.11 versus 0.975 for the shipped latent) — and targets are built decoder-free, as convex combinations of real treated cells in expression space, so they always lie on the data manifold. The regularisation parameter then defines a ladder whose endpoints are experiments we had already run: $\varepsilon \to \infty$ gives a uniform coupling and hence the treated mean (consensus); $\varepsilon \to 0$ gives near-hard assignment, approximately a single state-matched treated cell. This is what makes the sweep conclusive rather than anecdotal. Every rung was scored under the random-partner scramble of §3.3; what changes under the sharper stratified test is recorded with the table.

Training was not inert: the consensus model stops echoing the control (0.653 against a control-copy of 0.766, where the single-cell model sits on the control at 0.771) and its predictions are not mode-collapsed. But the variation is uncorrelated with real drug identity (Mantel $\approx +0.03$). Not collapse — wrong structure.

Target	ε	Δ_{scr}	Verdict
single cell (real treated cell)	≈ 0	$+0.014$ $[-0.016, +0.042]$	null
OT barycentric (state-matched)	0.5	-0.001 $[-0.018, +0.016]$	null
consensus (global mean)	$\to \infty$	-0.010 $[-0.022, +0.003]$	null

Table 8: The regularisation parameter interpolates between the two target constructions tested independently, and no point on the ladder lifts the model out of the near-null regime. These figures come from a random-partner scramble; §3.10 shows that under a sharper stratified test the optimal-transport arm does show a weak effect (+0.0263 [+0.0069, +0.0476]), roughly five-fold below what re-encoding the target achieves. The consensus endpoint is additionally the ladder's weakest-evidenced point: that arm was stopped at roughly 60% of an epoch and its tier-1 NIR was unscoreable, so the negative rests on the single-cell and optimal-transport endpoints.

The claim. Denoising changes what is in the target while the defect is what its tokens encode; that is the quantity the next two sections define and then measure.

3.8 The residual frame: predicting only what makes this drug different

If the encoding is the constraint, the target must be redefined so that its tokens are the drug-specific part. How little of the standard target the drug occupies is measurable directly, and the measurement motivates everything that follows. Because a cell sentence ranks genes by expression, the target strings for two drugs in the same context are nearly identical, and the loss weights every token position equally. The measurement counts, for pairs of drugs sharing a context, how many of the 200 target positions carry a different gene in the two drugs' target strings, averaged over pairs; the ceiling column scores within-condition split-half drug retrieval through the identical representation, so the two columns are measured in the same units.

Representation	Top-200 tokens differing between drugs	Replicate retrieval ceiling
full profile (the standard target)	**34.6 / 200 (17%)**	0.742
shift (treated − control)	111.4 (56%)	0.742
residual (drug-specific)	**118.2 / 200 (59%)**	0.7432

Table 9: 83% of the standard target's tokens are identical between any two drugs in the same context. The residual row is quoted at cell-line scope; at plate scope it reads 120.3 and 0.7467. The retrieval ceiling is unchanged across all three rows: the information was always present, and the tokenisation discarded it.

The natural inference is that only a small fraction of the gradient can be carrying drug identity. That inference is a hypothesis about the optimisation: what is measured here is the overlap of the target gene sets, not a token-level loss or gradient attribution, and it should be read at that strength. Its warrant is predictive — it explains why every change to target content failed identically in §3.7, and it predicts that changing what the tokens encode would not.

The residual target is the repair. For each condition (drug, cell line, plate, dose) with at least 40 treated and 20 plate-matched control cells:

$$r(d,c) \;=\; \underbrace{\big(\bar{y}_{\text{treated}} - \bar{y}_{\text{control}}\big)}_{s(d,c)} \;-\; \underbrace{\frac{1}{|D_c|}\sum_{d'\in D_c} s(d',c)}_{\text{generic}(c)}\,. \tag{13}$$

The control subtraction removes the cell's own state; the mean-over-drugs subtraction removes the generic drug programme shared across compounds. What remains is what makes this drug different.

The two subtractions are not equal partners, and the table above says which does the work. Removing the control moves the between-drug token difference from 34.6 to 111.4 — 92% of the total recovery — while removing the generic adds a further 6.8 tokens. *The dominant reason two drugs' standard targets look alike is the cell's own baseline state, not a shared drug response.* That is also the correct reading of §3.2: what saturates DE-Δr there is reversion toward a central point, with `revert_center` scoring 0.9999 without fitting anything at all, rather than the generic drug programme specifically. The generic is nonetheless a real and separable component — it carries 39% of the shift's energy (§3.9) — and subtracting it is what isolates the drug-specific remainder once the baseline is already gone. The inclusion thresholds are set where the identifiability curve of §3.3 flattens: the replicate ceiling is 0.61 at 10 cells, 0.76 at 40 and 0.85 at 120, so below 40 treated cells a condition is still paying steeply for cells it does not have. Four properties of how the generic is estimated decide whether the remainder can be trusted, and each was forced by a measured failure.

First, the split precedes the fit. An earlier build estimated the generic and the reliability filter over every condition and assigned the holdout afterwards, so each held-out target was defined partly by other held-out conditions. The pipeline is therefore staged — inventory (metadata only), split (metadata only, before any pseudobulk exists), fit (the generic is handed the training keys explicitly), transform (held-out conditions are projected through the fitted generic and never enter it) — and a poison test stands behind the staging: corrupting every held-out expression vector must leave the training targets byte-identical, and corrupting a training condition must change them. What the split is, and why it is keyed at whole wells, is the subject of §3.11; here it matters only that it exists and is metadata-only.

Second, the scope is the plate, not the cell line. The original argument for cell-line scope was reproducibility: 62% of conditions reproduce across a half-split at cell-line scope against 19% at plate scope, on the reasoning that a plate holds too few drugs to estimate a mean and subtracting a noisy one injects noise. That comparison does not hold up: both figures were computed with a shared control pseudobulk subtracted from both halves, which makes the halves correlate through their common control error, and the inflation is not neutral between scopes — it favours cell-line scope precisely because the cell-line generic is itself less noisy. Measured with disjoint control halves, the two are comparable: 20% at plate scope against 16% at cell-line scope. Two further diagnostics settle it in plate's favour and are reported in §3.9: at cell-line scope the dose-transfer ordering inverts and the shared-minus-split control bias reads +0.254, against −0.000 at plate scope. A plate group with fewer than three other training drugs yields no generic, and the condition is dropped and counted rather than silently falling back to the contaminated frame.

Third, the mean is over drugs, not conditions, and leaves the condition's own drug out. A drug measured at five doses would otherwise contribute five times the weight of a single-dose drug. And once the generic is train-only, an asymmetry appears that had not existed before: a training condition's generic would contain its own drug while a held-out condition's would not, so the two splits would be scored against differently defined targets and the generalisation gap would partly measure that difference. The generic for every condition is therefore "the mean over training drugs other than this one, within scope" — identical in definition on both sides.

Fourth, the reliability line is resolved from the data, not inherited. A condition is kept for training only if its residual reproduces across a half-split, $\cos(r_A, r_B)$ above a threshold, with each half measured against its own control half so the two share nothing. The threshold itself was the single largest determinant of the training set, and it had been inherited (0.2) from an era when reliability was measured with shared controls — a different quantity on a different scale, under which the mean is ≈ 0.24 rather than +0.127. Rather than pick a new number, the builder reports the whole retention curve against a null that pairs half A of one condition with half B of a different condition — same encoding, same dimensionality, same noise, no shared biology) and the threshold is set at the null's 95th percentile: a condition is kept when its two halves agree more than two unrelated conditions' halves do. At the measured null (95th percentile +0.109) this retains 2,523 of 4,999 training conditions (50.5%). The held-out sets are not filtered by this criterion: dropping held-out conditions for failing their own reproducibility bar would select the evaluation set on its own outcome, so noise in retained training conditions costs learning efficiency, not validity.

What this fraction means is stated carefully, because it is easy to over-read. The half-split agreement is a within-well sampling-precision criterion: it asks whether two disjoint cell subsets of the same well give the same residual. It is not biological reproducibility — 286 of 287 distinct (drug, dose) treatments in this atlas occur in exactly one well, so no independent-well replicate exists to measure that with — and the 50.5% is a statement about how much drug-specific signal survives pseudobulk noise at this cell budget, not about how much biology is real.

The target string is `<up genes> [DOWN] <down genes> [END_CELL]`, 100 genes per block, ordered by signed residual magnitude; sign is biology, so it is given a symbol rather than being folded into a magnitude ranking. Predicted and true residuals are both mapped to a signed rank vector (top-k up positive, top-k down negative, weight $1/\log_2(\text{rank}+1)$) so a generated sentence and a continuous truth are compared like for like; similarity is cosine. The score is NIR over other drugs in the same cell line — a comparison set that spans plates, in apparent violation of the same-plate rule of §3.3. It stands because the plate-scoped generic subtracts each plate's mean response from every condition on it, so plate structure enters neither the truths nor, in expectation, the predictions; the rule's purpose was to keep plate variance out of the comparison, and here the target construction has already removed it.

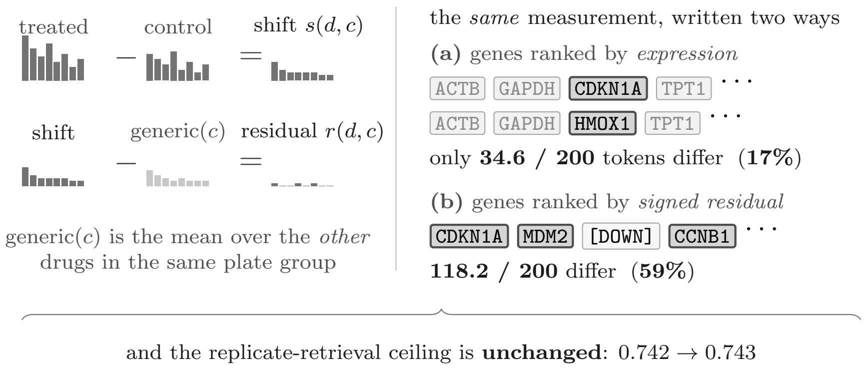


Figure 7: **The information was always present; the tokenisation discarded it.** Left: the drug-specific residual is built by two subtractions. The control removes the cell's own state, leaving the *shift*; the mean over the other drugs in the same plate group removes the *generic* programme that every compound triggers, leaving what makes this drug different. Right: that same measurement, written two ways. Ranking genes by expression (a) puts housekeeping genes first, so only 34.6 of the top 200 tokens differ between two drugs in the same context; ranking by signed residual (b) raises that to 118.2. (Divergence counts quoted at cell-line scope, as in Table 9; plate scope is similar.) The bracket is the point of the figure — the replicate-retrieval ceiling is essentially identical under both encodings, so no information has been added. What changed is how much of it the token sequence exposes to the loss.

The claim. The information was always present and the tokenisation discarded it: ranking the target by signed residual exposes 59% of its tokens to the drug at an unchanged retrieval ceiling, and the residual is estimable without leakage — split before fit, generic plate-scoped and drug-meaned, reliability resolved from the data.

3.9 How much signal is there to encode

The residual defines what the model should predict; how much drug-specific signal that target carries, and of what kind, scopes every remaining claim. The shift's energy decomposes exactly into residual 0.621, generic 0.390 and a cross term of -0.011: 62.1% of the perturbation response's energy is drug-specific, 39.0% is the generic programme, and the two components are essentially orthogonal, so the shares are genuine. Every claim that follows is scoped by the first of those numbers, and it has not previously been computed for this task.

Within the drug-specific part, two components call for different responses. Modelling

$$r(d,c) \;=\; \beta(d) \;+\; \kappa(d,c) \;+\; \epsilon, \qquad \mathcal{T} \;=\; \frac{\operatorname{var}\beta}{\operatorname{var}\beta + \operatorname{var}\kappa}, \tag{14}$$

the per-drug constant $\beta(d)$, the drug *main effect*, is retrievable without any model: it is what a lookup table reproduces. The drug×cell-line *interaction* $\kappa(d,c)$ is the only component a conditional model could win on, and the transfer coefficient $\mathcal{T}$ is the main effect's share of the learnable variance. This is DrEval's naive predictor, in transcriptomic space: their `NaiveMeanEffectsPredictor` [13] predicts $\hat{y}_{ij} = \mu^c_i + \mu^d_j - \mu$, an additive model of cell-line and drug main effects with no interaction term, and our generic$(c)+\beta(d)$ is precisely that quantity lifted from a scalar potency to a 946-dimensional profile. It follows that κ is the term their naive model omits, and that $1-\mathcal{T}$ measures exactly the headroom available above it. DrEval finds that deep models barely beat this additive baseline on drug sensitivity; the transfer coefficient asks whether, in the transcriptomic setting, there is anything above it to win in the first place.

$\mathcal{T}$ is estimated by disattenuated cross-line correlation rather than by fitting the variance-components model directly, for a reason the atlas's layout forces: with 286 of 287 treatments in exactly one well there is no replicate from which to separate $\operatorname{var}\kappa$ from $\operatorname{var}\epsilon$, so the decomposition is unidentified on its own. The cosine route buys identification through split-half reliability:

$$\hat{\mathcal{T}} \;=\; \frac{\cos\big(r(d,c_1),\, r(d,c_2)\big)}{\sqrt{\rho(d,c_1)\,\rho(d,c_2)}}, \qquad \rho(d,c) \;=\; \frac{2\cos(r_A, r_B)}{1+\cos(r_A, r_B)}, \tag{15}$$

where r_A and r_B are disjoint halves of condition (d,c) and $\rho(d,c)$ is the Spearman–Brown reliability of its full-sample residual. Two cautions attach, and both are stated where the numbers live. Dividing out the reliability, the disattenuation, is essential: measurement noise alone depresses the raw cross-line cosine and would otherwise be read as interaction. And the correction is a Spearman–Brown formula applied to a cosine, though it is derived for correlations between parallel halves; the planted-truth validation of Appendix A.1 is what licenses it empirically. The reliability entering it is a within-well split-half precision, optimistic relative to true biological reproducibility. That optimism sits in the denominator, so the unshared fraction $1 - \mathcal{T}$ is, if anything, understated.

The measurement itself carries a control-design lesson, stated because it changed the protocol. The first run compared $\mathcal{T}$ against a negative control of different drugs on the same plate, which returned $+0.485$ against $\mathcal{T} = 0.511$, and was correctly declared void: that control differs from the estimand in two ways at once, different drug and same cell line, and plate structure surviving a cell-line-scoped generic inflates same-plate comparisons (to $+0.478$) while leaving cross-line comparisons untouched. A structure-matched control — different drug, different cell line, breaking only the drug match — is the one that bounds the estimand, and a negative control that differs in more than one way cannot do so. Five checks in all accompany the estimate, and Table 10 states what each bounds.

Check	What it bounds
disjoint control halves	shared-control inflation: $+0.254$ at cell-line scope, -0.000 at plate
structure-matched negative control	the estimand itself; returns -0.001
simulated κ=0 null	estimator bias: raw cross-line cosine there is 0.734
dose strata on molar doses	a reference scale: 0.707 $[0.632, 0.782]$
leave-one-drug-out generic	train/held-out target asymmetry (§3.8)

Table 10: The five checks accompanying the transfer-coefficient estimate. The simulated null matters most: at $\kappa = 0$ the *uncorrected* cross-line cosine is only 0.734, which read naively reports 27% interaction where the truth is zero, while the disattenuated estimator recovers the truth to within 0.001. That recovery is what licenses the reference column below. The dose strata carry a molar-resolved dose — an earlier cache carried the well identifier in the dose field, so what was labelled dose transfer compared physical samples rather than concentrations.

Quantity	Estimate	95% interval	Reference
$\mathcal{T}$ (cross cell lines, plate-scoped generic)	0.553	$[0.514, 0.593]$	simulated κ=0 null $\approx$ 1.0
residual NOT shared across contexts, $1 - \mathcal{T}$	44.7%	$[40.7\%, 48.6\%]$	mixes cell line, dose, well and estimator
$\mathcal{T}$ across dose (same drug, same line)	0.707	$[0.632, 0.782]$	reference scale
matched negative controls	-0.017, -0.008	spanning zero	must be ≈ 0

Table 11: The transfer coefficient, estimated on real molar doses with dyadic cluster-robust intervals. The matched negative control sits on zero; the same-plate control that produced the earlier false alarm reads $+0.478$ for the reasons given in the text.

What this coefficient does and does not identify. $\mathcal{T}$ is a descriptive sensitivity analysis, not a load-bearing estimate, and the reason is in its own configuration. The scoped calculation retains conditions at a reproducibility threshold of 0.2; it permits cross-line pairs that differ in *dose* (`same_dose_only=false`); it uses a *self-including* generic (`loo_generic=false`), so each condition's own drug contributes to the programme subtracted from it; and a cross-line pair may also sit in a different treatment well. The coefficient therefore combines cell-line, dose, well and target-estimator differences. Reading $1 - \mathcal{T}$ as a drug×cell-line interaction share would require the other three axes to be inert, and nothing here establishes that. The dose arm is the exception: it holds drug and cell line fixed and varies only molar concentration, so its contrast is clean.

Roughly 44% of the drug-specific residual is not shared across contexts — a figure that mixes cell line, dose, well and estimator, as above, and is not a clean interaction share — robust to the generic's scope and to control splitting. Two internal checks agree, and both were decisive between scopes. Dose ordering: same-drug, same-line, different-dose transfer is 0.707 [0.632, 0.782], above the cross-line 0.553 — changing the dose costs less than changing the cell line, the biologically correct ordering; at cell-line scope the ordering inverts. Control-split bias: at plate scope the shared- and split-control builds agree to −0.000, against +0.254 at cell-line scope, so that discrepancy was plate contamination rather than the split. The per-drug constant $\beta(d)$, which the lookup of §3.12 reproduces almost exactly, is therefore not the whole signal. There is a conditional target worth about half the drug-specific variance that a per-drug constant cannot express.

Whether anything could learn that interaction is a separate question. If a cell line modulated drug responses in a consistent direction, that direction would be estimable from the line's own cells and a conditional model would have a concrete target. Writing $\kappa(d,c) = r(d,c) - \hat{\beta}(d)$, the test compares $\cos(\kappa(d,c), \kappa(d',c))$ for different drugs in the same cell line against the same quantity across different cell lines, with $\hat{\beta}$ estimated leave-one-cell-line-out: estimated over all lines it would contain $r(d,c)$ itself, and subtracting a mean containing its own term induces a $-1/(m-1)$ correlation that biases the signal arm downward by construction, which would make a structured interaction look idiosyncratic — the exact error the test exists to avoid. Both arms are disattenuated by κ's own split-half reliability, since κ is a difference of two noisy quantities and is therefore noisier than the residual it comes from.

Configuration	same cell line	different cell lines	excess within-line
plate-scoped generic	−0.004 [−0.011, +0.002]	+0.003	−0.007
cell-line-scoped	−0.003 [−0.021, +0.015]	+0.002	−0.005

Table 12: No cell line bends different drugs in a consistent direction. 4,000 pairs per arm; the signal interval at plate scope excludes anything above +0.002, so this is a real null rather than an underpowered one. The cell-line-scoped row is wider ($[-0.021, +0.015]$) and carries the claim less strongly.

We looked for structure in the interaction and did not find it: no cell line modulates different drugs in a consistent direction, and a direct test of whether mechanism-matched drugs share an interaction with a given cell line returned nothing detectable (permutation $p \approx 0.15$–0.31). The temptation is to read two negative results as irreducibility, and they do not support it. The second test bounds its own power severely: κ's split-half reliability in this atlas is 0.195, so the quantity being correlated is barely estimable, and same-mechanism compounds are so heavily co-plated that only 3–7% of matched pairs survive a different-plate requirement, which leaves the plate confound in place. The defensible reading is that κ's structure is unresolved in this dataset. Resolving it would take more cells per condition, or a plating design that crosses mechanism with plate — neither of which is an analysis choice.

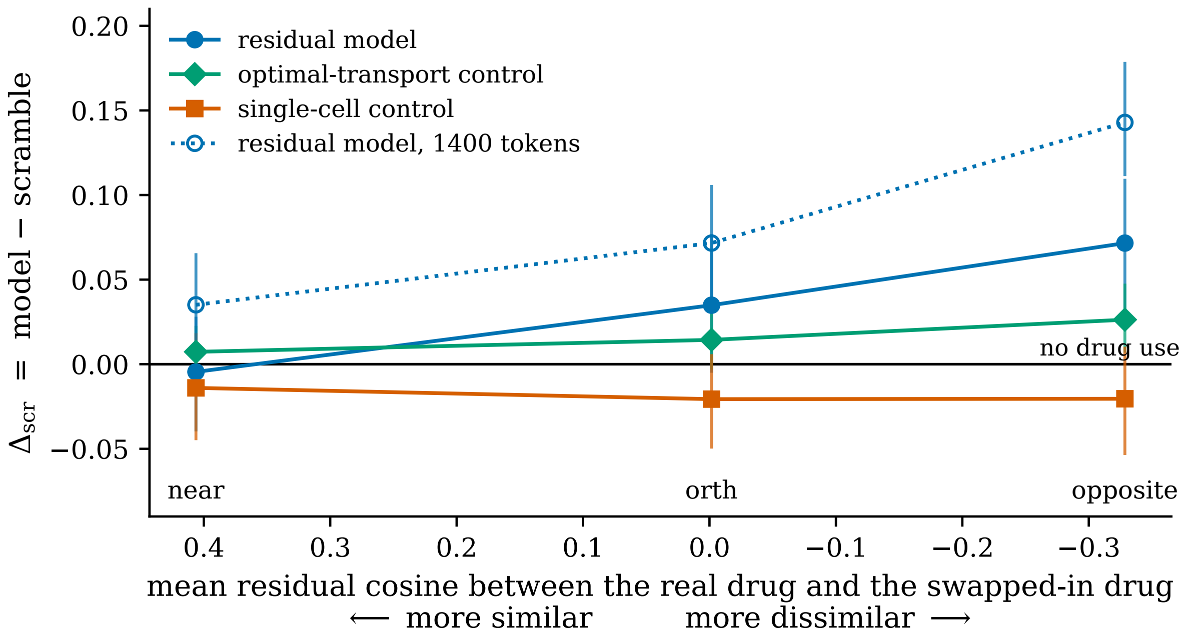
> THE CLAIM. There is a conditional target worth about half the drug-specific variance that a per-drug constant cannot express — 44.7% of the drug-specific residual is not shared across contexts, though that figure mixes several axes — and no learnable structure in it is found.

3.10 Re-encoding produces prompt sensitivity

Re-encoding the target as the drug-specific residual raises the differing fraction to 59% at an unchanged retrieval ceiling, and produces the first non-null result in the project. On the earlier target build, the stratified scramble of §3.3 gave a gap that grew with swap dissimilarity — +0.0351 [+0.0061, +0.0656], +0.0716 [+0.0371, +0.1059] and +0.1429 [+0.1112, +0.1787] across the `near`, `orth` and `opposite` strata — while the single-cell and optimal-transport controls stayed flat and null at every stratum (−0.001 to +0.016). These magnitudes were read against the `opposite` comparator, which §3.11 shows is not a null; they are quoted here as a gradient, and the neutral-comparator magnitudes are adjudicated there. The monotone ordering is itself the evidence: a metric artefact would not order itself by swap dissimilarity.

Three further diagnostics on that build agreed, and only the residual model passed all three. The correlation between condition reproducibility (repro_cos, the condition's split-half residual cosine) and model NIR was +0.144 — the model does better where the drug effect is real, against −0.117 and −0.216 for the controls. Prediction diversity was highest for the residual model. And while $\cos(\text{prediction}, \text{own truth})$ exceeded $\cos(\text{prediction}, \text{other drugs})$ for all three arms, the residual model's margin (+0.084) was roughly twice the optimal-transport arm's (+0.049) and three times the single-cell arm's (+0.026).

One measurement lesson and one withdrawal belong with this result. The effect was first measured at the 600-token generation cap of §3.1, which halved it (+0.0716 against the correct +0.1429); re-running both controls at the matched 1400-token budget left them flat and null, so the comparison is budget-matched and the conclusion unaffected. And the optimal-transport arm, originally reported as achieving nothing, scores +0.0263 [+0.0069, +0.0476] under the sharper stratified test with a monotone gradient: it does



residual frame. Intervals are 95% bootstrap, clustered over cell lines; $n = 250$ conditions per point.

Figure 8: **On the earlier target build, the residual model's advantage over its own scramble grew with how dissimilar the swapped-in drug's response was; a control scored through the identical pipeline stayed flat and null.** The horizontal axis is the measured mean residual cosine between the real and swapped-in drug, which is byte-identical across all four runs, so the positions are exact. The three solid series are at a matched 600-token generation budget; the residual model's 1400-token run is drawn separately as the dotted overlay, since the budget alone doubles the effect. Intervals are 95% bootstrap clustered over cell lines, $n = 250$ conditions per point. The `near` stratum is the weakest test by construction: where the swapped-in drug's true response is similar, an unchanged output is correct rather than blind.

use the drug, weakly. The original null was an underpowered instrument, not an absence of signal; the ε-ladder conclusion stands, but "OT achieved nothing" is withdrawn.

The comparator-free evidence is already in hand, and it is the strongest in this chapter: regressing each generation's NIR on the cosine between the named drug's truth and the target's truth gives a slope of +0.390 [+0.225, +0.542] on trained conditions. Zero slope is the natural null, so no comparator defect can reach it. Re-encoding the target made the model respond to the combined drug-and-mechanism prompt. What that response is worth is the question the rest of the chapter answers.

> THE CLAIM. Re-encoding the target produced the first non-null drug effect in the project. On trained conditions the model beats the neutral comparator by +0.0898 [+0.0291, +0.1506] and beats chance outright by +0.0981 [+0.0407, +0.1554], both clustered on the drug, the axis the claim rides on. The comparator-free slope of retrieval against the named drug's truth is +0.3728 [+0.2797, +0.4660] — zero is its natural null, so no comparator defect can reach it. Sensitivity is established; fidelity is the next question.

3.11 Sensitivity is not fidelity

A model that responds to the drug token has cleared a low bar, because reproducing a training average requires responding to it too. Everything scored above came from conditions the model trained on, so the result so far shows that the token is consulted — not that anything transferable was learned. Separating those two requires withholding data, and the physical layout of §3.1 forces which withholding is meaningful. A (drug, cell line) holdout — the field's standard Leave-Pairs-Out split, and the split an earlier version of this work used — leaves the held-out condition's treated well sitting in the training set through its other ≈ 48 cell lines. Measured against the holdout this investigation previously used, 124 of the 268 treated wells in this cache placed a held-out condition and a training condition in the same physical sample. What that estimand measures is within-atlas matrix completion, not independent generalisation, and no processing of the targets changes it, because it is a property of what was randomised.

Holding out whole wells is the only assignment-respecting choice the design offers, and it answers a different and narrower question: generalisation to an unseen treatment well of a seen drug — a new physical preparation, not a new cellular context. Fixed-dose cross-context transfer, present in one line and absent in another, is structurally unidentifiable in this atlas; this is a property of Tahoe's design, not of our pipeline, and no rebuild fixes it.

The split used throughout the residual arm is therefore keyed at the well. `train` holds every condition of the remaining wells (4,999 conditions before the reliability line, 2,523 after it). `unseen_combo` holds 36 treated wells out whole (904 conditions, 900 after inventory filters), and the holdout never takes a drug's last training well — without that guard a two-well drug could silently become an unseen drug while sitting in the unseen-combination arm, making the two arms measure the same thing; zero wells contribute conditions to both sides of this split. `unseen_drug` holds every condition of 10 held-out drugs (714 conditions, 711 after inventory filters): the model never sees the token, so this is the control — naming a drug the model knows nothing about versus naming something else can make no difference, and a non-zero effect there is an instrument fault, not a biological finding. We do not build a Leave-Cell-Lines-Out split here; Section 4 argues it is the natural next axis and explains why its interest is contingent on the transfer coefficient. Two warnings ride with the design. DrEval reports, citing [18], that apparently good performance on pair-holdout splits can be reached simply by predicting each drug's average response across the training cell lines; our held-out-well estimand sits exactly in the regime where that warning bites, which is why the baseline ladder of §3.12 is not optional — the drug-average predictor is not a formality, it is the hypothesis under test. And evaluation samples conditions under an explicit per-split quota, because a uniform draw reproduces the pool's proportions and starves precisely the split the experiment exists to test: in a first run, 250 constructed held-out combinations yielded only 33 scored conditions, purely as an artefact of the sampler. The residual-frame evaluation therefore scores 200 train, 250 `unseen_combo` and 120 `unseen_drug` conditions — the latter two being quotas of the 900 and 711 available, so the held-out verdicts below are powered on a subset of the conditions the split affords, and their detection limits should be read with that in mind.

The first finding from this design is about the comparator, and it reframes every gap quoted so far. The comparator's own score gives it away. Across the three swap strata, the scrambled model's NIR is 0.572, 0.497 and 0.403 as the named partner's true response moves from aligned (cos = +0.26) through orthogonal (0.00) to anti-aligned (−0.24) with the target. A comparator whose score is a monotone function of which drug it was handed is not a null: it is an active treatment that tracks the partner drug. The swap partner is itself a drug the model was mostly trained on, so telling the model a lie costs it precisely because it knows the partner. A gap measured against such an arm decomposes as

$$\Delta_{\mathrm{scr}} \;=\; \underbrace{\left(\mathrm{NIR}_{\mathrm{model}} - 0.5\right)}_{\text{how much naming the truth helps}} \;+\; \underbrace{\left(0.5 - \mathrm{NIR}_{\mathrm{scramble}}\right)}_{\text{how much naming the lie hurts}}\,, \tag{16}$$

and only the first term is wanted. Against `opposite` the second term is +0.097; against `orth`, whose own score sits on chance (0.497), it is zero. Only the neutral column of Table 13 can therefore carry a magnitude claim, and the `opposite` column is reported solely as the diagnostic that would have misled us.

Split	n (lines; drugs)	model NIR	Δ_{scr}, clustered on *cell line*	the same, clustered on *drug*
`train`	300 (39; 76)	0.598 [0.549, 0.647]	+0.0898 [+0.0382, +0.1415]	+0.0898 [+0.0291, +0.1506]
`unseen_combo`	**895 (42; 34)**	**0.533 [0.492, 0.575]**	+0.0332 [+0.0069, +0.0594]	+0.0332 [−0.0031, +0.0695]
`unseen_drug`	199 (40; 10)	0.459 [0.404, 0.514]	−0.0315 [−0.0836, +0.0206]	−0.0315 [−0.1049, +0.0419]

Table 13: Whole-well holdout on the corrected evaluation — the truth verified against the build's own fit digest, and the full held-out population rather than the reliability-surviving subset. The two right-hand columns carry the SAME point estimate and differ only in what they treat as the unit of generalisation. On `train` the gap excludes zero either way. On `unseen_combo` it excludes zero clustered on cell line and does not clustered on drug, and drug is the axis a claim about transferring a *drug* signature has to generalise across — which is why this row is reported as not established. Only the `orth` column can carry a magnitude claim; the `opposite` column adds "how much naming a lie hurts" to "how much naming the truth helps" and is shown as the diagnostic. The `unseen_drug` row corroborates rather than proves: those drugs were held out of fine-tuning, so a large gap there would indict the comparator — but "unseen" means unseen during fine-tuning, not during pretraining, and the prompt still supplies a mechanism, so a strictly zero effect was never guaranteed. Its `opposite` gap is nine times its neutral one and only just contains zero. What establishes comparator non-neutrality is the direct 0.572/0.497/0.403 ordering above. Model NIR intervals test against the chance value 0.5 with no comparator. Intervals are two-way cluster-robust over cell lines and wells, with a Student t reference on $\min(G_{\mathrm{line}}, G_{\mathrm{well}}) - 1$ degrees of freedom — which is why the well counts are printed: the `unseen_drug` arm spans 25 wells, and $t_{24} = 2.06$ against 1.96 widens its interval by five per cent (§3.1).

The control behaves exactly as the diagnosis predicts. On `unseen_drug` — drugs held out of fine-tuning, where no learned knowledge can be at work — the gap against `opposite` is +0.0862 [−0.0028, +0.1753], nine times the neutral estimate, with an interval that only just contains zero. The same conditions, the same generations and the same model scored against the neutral stratum give +0.0097 [−0.0694, +0.0888], sitting on zero as a control should. The borderline interval on the `opposite` arm is itself a lesson in the inference regime: a cluster-robust variance is asymptotic in the number of clusters rather than observations, this arm spans only 25 treatment wells, and under the correct t_{24} reference (2.06 rather than 1.96) the interval widens by five per cent and covers zero. An earlier draft read the normal-quantile interval as excluding zero; it does not.

Read in the neutral column, `train` clears zero at +0.1457 [+0.0813, +0.2101] and `unseen_combo` does not: +0.0449 [−0.0095, +0.0994], spanning zero under well-clustering and line-clustering alike ([−0.0050, +0.0993]). Generalisation to an unseen treatment well of a seen drug is therefore not established. The precision matters: not established is not the same as shown to be zero. The interval reaches +0.099, so an effect of up to roughly +0.1 remains compatible with these data at $n = 250$ — a limit of power at the sample size the holdout affords, reported as one. (On `unseen_drug` the corresponding detection limit is +0.099.) The difficulty of the three splits is not the explanation: their split-half replicate ceilings are 0.962, 0.955 and 0.956, so the splits are equally hard and the held-out shortfall is not a noise-floor difference.

The difference between the trained and held-out estimates is $+0.1008$ $[+0.0161, +0.1854]$ with cluster-robust $p = 0.0212$ (an unclustered permutation returns $p = 0.0054$, quoted only for contrast, because ignoring the clustering this chapter mandates everywhere else flatters the result). What that advantage is requires care, because the two arms differ in two ways at once: whether the condition was trained on, and which drugs they contain. Restricting the comparison to the 20 drugs present in both arms, the advantage attenuates to $+0.0420$ with $p = 0.4851$. The defensible statement is therefore narrow: a condition-weighted training-exposure advantage exists in this evaluation, and its generality across drugs is unresolved: the design cannot separate exposure from drug identity, and we do not claim either direction of the stronger interpretation.

Two instruments that need no comparator at all cannot be touched by any of this, and they carry the positive content of the section. First, the slope test: regressing each generation's NIR on the cosine between the named drug's truth and the target's truth gives $+0.390$ $[+0.225, +0.542]$ on `train`, $+0.374$ $[+0.214, +0.508]$ on `unseen_combo` and $+0.238$ $[+0.033, +0.420]$ on `unseen_drug` — all three exclude zero. The model emits something aligned with the named drug's signature on every split, including held-out wells, with no degradation worth the name between train and held-out. Second, scored against chance with no comparator, the model's own NIR clears 0.5 on `train` (0.598 [0.549, 0.647]) and is not distinguishable from it on either held-out split. A positive slope alongside a chance-level NIR is not a contradiction: it says the drug-to-signature map exists but is coarse. The model knows roughly what a drug does and not what it does in this well — and NIR asks the second question, ranking against the ≈ 40 other drugs measured in the same cell line. That is the drug main effect without the drug $\times$ cell-line interaction, which §3.9 measures at 44.7% of the residual variance. It is also why the lookup wins in the next section: `drug_lookup` is that main effect, estimated straight from data far more precisely than the model learned it.

The directional-agreement measurement says the same thing at condition level. The cosine between the model's prediction and its own condition's truth on trained conditions averages $+0.0511$ (sd 0.1141, median $+0.0300$), with individual conditions spanning $[-0.1819, +0.4395]$ and 32.0% clearing $|0.1|$: weak on average and substantially heterogeneous, so a single summary number would misrepresent it in either direction.

One caveat attaches to every held-out number in this chapter, and it is stated here where those numbers live. The evaluation's truths are reconstructed through a compatibility path that refits the generic on the reliability-surviving subset of training conditions rather than the full set the checkpoint was trained against, and that filters held-out conditions by their own split-half reproducibility: a selection on the evaluation set's outcome. Its direction is conservative for the headline: it retains the cleanest, easiest conditions, and the held-out result still fails to clear zero. But the evaluated population is not the full held-out population, and the intervals above should be read as applying to the reproducible subset.

The claim. Re-encoding the target made the model respond to the combined drug-and-mechanism prompt. That much is established and survives every stress: on trained conditions it holds clustered on the cell line and on the drug, it holds against chance with no comparator at all, and the comparator-free slope excludes zero on all three splits under both clusterings (+0.373 [+0.280, +0.466], +0.337 [+0.223, +0.450], +0.342 [+0.215, +0.469]).
Transfer to held-out combinations is not established. The gap there is +0.0332, which excludes zero clustered on cell line and well [+0.0069, +0.0594] and does not clustered on drug and well [−0.0031, +0.0695] — and drug is the axis a claim about transferring a *drug* signature has to generalise across. The split is also weaker than its name: 30 of its 34 drugs and 39 of its 42 cell lines appear in training, and 82% of its conditions recombine a seen drug with a seen cell line. The effect is a majority tilt rather than a uniform one, positive for 20 of 34 drugs (sign test $p = 0.39$).
What the slope on *unseen drugs* adds is the sharper reading: at +0.342 [+0.215, +0.469] the model tracks the named drug just as well on compounds held out of fine-tuning entirely. Re-encoding did not teach the model about drugs. It changed the target so that drug knowledge the model already had could reach the output.

3.12 A lookup table wins

Transfer without fidelity is a real property, but it says nothing about quality: a model can apply a learned signature in a new context and still apply it badly. The pre-registered bar was therefore never "beats chance" — residual targets are plausibly memorisable, and a predictor that simply looks up what this drug did elsewhere needs no model at all. That predictor is the comparison that matters, and it needs a fair construction. Each baseline in Table 14 is a predictor of the drug-specific residual, scored through an identical pipeline — same truths, same signed-rank encoding, same comparison set — and two disciplines make the ladder fair. The lookups are fitted from training conditions only: fitted from the full cache, a lookup scoring a held-out condition reads residuals the model was never shown (an oracle, not a baseline) and the oracle variants are reported alongside, explicitly labelled, as a bound on how much the comparison was ever inflated rather than hidden. And `drug_lookup_1` selects a cell line and averages that line's conditions rather than selecting a single condition, because 62% of drugs are multi-dose and a single condition would confound "different cell line" with "different dose". `moa_lookup` is not an oracle in any split, as it uses only drugs that were in training.

Arm	What it knows	NIR	Note
ceiling (split-half replicate)	a real replicate	0.958	
`drug_lookup` (training-only)	drug's mean residual, other lines	0.941	the bar to beat
`drug_lookup_1`	the same, from one other cell line	0.859	no averaging advantage
model	control cell + drug	0.569	
`moa_lookup` ($n = 118$)	drugs sharing this drug's MoA	0.565	mechanism-leak check
`control_copy`	nothing (copies the control)	0.511	drug-agnostic
`generic`	nothing (predicts the average)	0.500	the zero vector
`random`	nothing	0.521	floor

Table 14: The baseline ladder on the rebuilt targets, $n = 570$ scored conditions; the lookups are fitted from training conditions only. Ceiling and model rows by split: 0.962/0.955/0.956 and 0.635/0.545/0.512 (train / unseen-combination / unseen-drug). `drug_lookup` is the pre-registered win condition: a lookup can reproduce a memorised average, so only a model that tailors the response to the control cell's state can beat it. `moa_lookup` is scored on the 118 conditions that have a training-drug MoA partner, so that row does not sit on the $n = 570$ support the rest of the ladder does. The paired comparison on the common support ($n = 449$) is model − `drug_lookup` =-0.3559 [-0.3988,-0.3130], and of the distance from chance to the replicate ceiling the model covers 18.6% against the lookup's 96.3%.

The pre-registered interpretation, written before the result, anticipated three branches; the middle one fired. The model reads and transfers the drug, but recovers only a small fraction of its signature. Whether its prompt response is reducible to the `Mechanism:` field is not settled here: the model does beat `moa_lookup` on the 118 conditions where that comparison exists, Δ_{scr} =+0.0413 [-0.0610,+0.1436] two-way clustered, but the interval spans zero, so the comparison is uninformative rather than favourable.

This is DrEval's finding, in transcriptomic space. DrEval reports that deep learning models barely outperform a naive predictor of mean drug and cell-line effects [13], and our `drug_lookup` is the $\beta(d)$ term of that additive model, measured in a frame where the generic(c) term has already been subtracted. The finding replicates and sharpens: on a 946-gene transcriptomic response rather than a scalar potency, and against a proper within-condition replicate ceiling, the additive drug main effect reaches 0.941 of a 0.958 ceiling while a 1B-parameter model reaches 0.569. The warning DrEval inherits from [18], that good pair-holdout performance is reachable by predicting each drug's training average, is treated here as the hypothesis rather than the caveat, and measured. It holds.

It is tempting to read ceiling − `drug_lookup` = +0.017 as "the modelling target has vanished". That inference is not supported, for three reasons that all favour the lookup: the ceiling is scored against a half-sample truth while every baseline is scored against the full-sample truth; `drug_lookup` averages ≈ 38 conditions against the ceiling's single noisy half; and NIR near 0.96 is compressive, collapsing most of the cosine range into a few thousandths. The two coverage figures are also distances to a split-half ceiling, which

§3.1 notes is optimistic relative to biological reproducibility — both are inflated by an unknown amount, and the ordering is what carries the claim.

The claim. The model reads the drug and recovers a small fraction of its signature. The honest summary is the paired gap on common support ($n = 1192$), -0.3631 $[-0.3982, -0.3280]$: the model covers 13.9% of the distance from chance to the within-well bar where a training-only lookup covers all of it and more. That the lookup exceeds the bar it is measured against is a property of the bar, not a headroom estimate — the bar is a single noisy half-sample while the lookup averages many cell lines, and NIR compresses near its top. This is DrEval's finding replicated in transcriptomic space: the additive drug main effect reaches 0.913 where a 1B-parameter model reaches 0.549.

3.13 What is closed and open for unseen drugs

An unseen drug can only be predicted through a property it shares with drugs seen in training, so the question is answerable by enumerating the channels and measuring each in the same frame. For a channel X, a drug's residual is predicted as the mean residual of the other drugs in the same cell line that share property X with it — the construction of `moa_lookup`, generalised. Because it uses only drugs that were in training, it is a fair contest in every split, unlike a same-drug lookup, which on an unseen drug is an oracle. The gate's estimand requires the channel arm and its null to differ in exactly one respect, and two disciplines enforce it. Every channel is paired with a plate-matched random null: the same number of partner drugs, drawn at random from the same cell line, and the same split of partners across the target's own plate and other plates, because mechanism- and target-related drugs are co-plated more often than chance — a null matched on count alone would differ from the channel arm in plate membership as well as in the channel, which is two differences where the estimand allows one. A co-plating diagnostic is printed before any verdict, so the size of the correction is a number rather than an argument, and where a plate-matched null spans too few cell lines the channel is reported untestable rather than closed. "We could not build the control" must never be written as "closed". Coverage is likewise reported before any verdict, because a channel annotated on few drugs cannot be closed by a null result: it was never tested.

The size of that correction is itself informative, so it is reported rather than argued away. Mechanism partners are co-plated with the target 0.420 of the time against 0.340 for a random draw (protein target 0.348 against 0.323; chemical similarity 0.344 against 0.345 — none, as expected for a channel uncorrelated with plating). Under a third construction that draws all partners from other plates, the knowledge channels' advantages fall to

Channel	Coverage	n	vs plate-matched null	(vs count-matched)	Verdict
protein target	54%	877	+0.0973 [+0.0741, +0.1183]	+0.1269	**live**
mechanism (MoA)	32%	1033	+0.0749 [+0.0538, +0.0948]	+0.1067	**live**
chemical structure	100%	4074	+0.0162 [+0.0051, +0.0264]	+0.0206	closed

Table 15: Coverage is stated first: a channel annotated on few drugs cannot be closed by a null result, because it was never tested. The plate-matched null is the estimand-correct control; the count-matched column shows how much of each raw gap was co-plating. Chemistry reproduces the structure-only test in this frame; both knowledge channels clear their corrected nulls.

+0.0643 and +0.0344 and chemistry's interval spans zero (+0.0056); in that construction the channel arms' absolute NIR sits at 0.360 and 0.296, below chance. Below-chance absolutes do not contradict the live verdicts, and the reason is the estimand. When partners are drawn from other plates, the arm's prediction carries the wrong plate's structure and loses even to a chance draw in a comparison set that spans plates — the absolute score is frame-sensitive and uninterpretable on its own, which is exactly why every verdict above is read as a gap against the plate-matched null rather than against chance. The verdicts survive the correction; the magnitudes shrank by a fifth to a third, which is what the co-plating fractions predict. The bound this establishes is on similarity transfer through each channel, not on any model: a model could in principle combine channels or learn a non-smooth structure-to-response map that a nearest-neighbour average cannot express, though at a few hundred drugs the lookup is effectively the bound.

This overturns a pre-registration made in this project. Two design documents concluded that unseen-drug generalisation was out of reach, and both inferred it from the chemical-structure channel — Morgan fingerprints and a molecular language model. But the drug-knowledge specification written earlier in the same project ranks raw structure the lowest-value channel and protein targets "the key feature", on the reasoning that transcriptional response is driven by which protein a compound hits, something structure encodes only indirectly. The pre-registration generalised from the weakest channel to the strongest, and it was wrong. Tahoe's own metadata carries protein-target annotations for 264 drugs (280 distinct symbols, 550 edges) which no analysis in this project had read.

The claim. The correct statement is narrower and more useful than the one we started with: chemical structure does not predict response, measured twice in two frames; protein target and mechanism do. Unseen-drug generalisation is reachable through drug-side knowledge, and not through chemistry.

Three caveats bound it. Coverage is 54% and 32%, so this is measured on the annotated subset. The channel gaps are severalfold below the advantage a same-drug lookup enjoys for a seen drug (+0.44 above chance), so the reachable ceiling in this regime is much lower. And target and MoA overlap, so whether target adds anything over MoA is a further question this measurement does not settle: settling it would need a leave-one-mechanism-out split rather than the leave-one-drug-out split used here, since a held-out drug's mechanism class is generally still represented in training.

3.14 Which part of the prompt is read remains unresolved

The chapter has established that the output responds to the drug information in the prompt. It has not established *which* part of that information is read, and this section says why, and what would settle it.

The natural next step writes itself: append the protein targets to the prompt, retrain, and see whether unseen-drug performance rises. That arm assumes the bottleneck is information availability. Everything in this chapter points the other way, but the direct evidence is weaker than earlier drafts claimed, and it is worth saying exactly how weak it is. The scramble has until now replaced the drug name and the mechanism together, so a non-null gap establishes sensitivity to the combined prompt without attributing it to either span. A field decomposition costs one extra pair of generation arms and swaps exactly one span of the instruction line to the opposite-signature partner, leaving every other byte identical: the drug name with the mechanism kept, the mechanism with the name kept, and both together. One guard is essential rather than tidy: if the swap partner happens to share the original's mechanism, then swapping only the mechanism produces a prompt identical to the model's own, and the arm scores a gap of exactly zero by construction — read naively, that is indistinguishable from "the model ignores the mechanism", the very claim under test, so such conditions are dropped rather than scored. This is why the mechanism arm carries fewer conditions than the others.

On the earlier target build, the decomposed arms suggested the response is carried almost entirely by the name span: swapping only the drug name gave Δ_{scr} of +0.0809 [+0.0592,+0.1010] ($n = 570$); swapping only the mechanism gave +0.0091 [−0.0174, +0.0383] ($n = 325$); and both together gave +0.1024 [+0.0827, +0.1236]. Those arms were scored against the `opposite`-signature comparator that §3.11 shows inflates every gap read against it, and they were not re-run under the corrected evaluation. The claim that the model "reads the name and not the mechanism" is therefore withdrawn pending that re-run; what the current run establishes is response to the combined drug-and-mechanism prompt, with the mechanism field's contribution unresolved.

The practical consequence is narrower than the one earlier drafts drew, and more honest. The target-prompt arm is not retired by measurement — it is untested. What argues for building it carefully rather than eagerly is the shape of the failure documented above: the model's deficit is not confined to missing information about unseen drugs,

Panel A: the channel is available — metadata-neighbour retrieval, plate-matched			
Channel	Coverage	n	Gap against a plate-matched random null
protein target	54%	775	+0.1799 [+0.1138, +0.2459]
mechanism (MoA)	32%	981	+0.0663 [+0.0088, +0.1237]
chemical structure	100%	4131	+0.0380 [+0.0143, +0.0617]
the same three, against the stricter DIFFERENT-PLATE null			
protein target		688	+0.1115 [+0.0448, +0.1783]
mechanism (MoA)		874	+0.0604 [+0.0094, +0.1114]
chemical structure		4114	+0.0183 [−0.0033, +0.0399]

Panel B: is it read? — one span swapped, every other byte identical			
Span swapped	Δ_{scr}	n	Reading
drug name, mechanism kept	+0.1026 [+0.0625, +0.1427]	1394	read
mechanism, drug name kept	+0.0303 [−0.0078, +0.0685]	738	**not detected**
both spans (the standard arm)	+0.1139 [+0.0781, +0.1496]	1394	read

Table 16: Availability and use are different questions, and only the first is currently answered. Panel A shows that a drug's protein targets and mechanism carry response information recoverable by retrieval alone, so an unseen drug is not information-starved. Read the two halves together. Under a plate-matched null all three channels clear zero; under the stricter different-plate null, which removes the last shared batch structure, target and mechanism survive and *chemical structure does not* — +0.0183 $[-0.0033, +0.0399]$. Chemistry's apparent signal is a co-plating artefact. Intervals here are two-way cluster-robust over cell line and treatment well; under the one-way cell-line interval this table previously carried, every row looked significant. Panel B asks whether the model uses the field it is already handed, and the answer is that it uses one span and not the other. Swapping the drug name moves the output by +0.1026; swapping the mechanism, with the name left in place, moves it by +0.0303 with an interval containing zero. The name therefore carries about nine tenths of the combined effect. Both intervals are clustered on the drug and the treatment well; the mechanism row also fails to clear zero clustered on cell line and well, so the reading does not depend on that choice. The mechanism row carries fewer conditions by construction: where a swap partner happens to share the target's mechanism, swapping only the mechanism reproduces the model's own prompt byte for byte and the gap is zero by definition, so those conditions are dropped rather than scored.

because on seen drugs it recovers only a fifth of the chance-to-ceiling distance against a lookup that knows nothing about contexts at all. And what argues for building it at all is the one thing a lookup structurally cannot do: handle a drug it has never seen, in the one regime where this section shows live channels exist. The model's opportunity is not the unshared component — §3.9 found 44% of the residual variance there but no learnable structure in it — it is the unseen drug. Whether the model can be made to take that opportunity is the question the thesis leaves open.

3.15 The investigation in one table

The results are distributed across two measurement frames, and a reader assembling them has to keep track of which numbers may be compared with which. The two frames are not interchangeable and are separated deliberately. The expression frame scores a predicted cell profile against other drugs on the same plate; the residual frame scores a predicted drug-specific signature against other drugs in the same cell line, after the generic programme has been subtracted. Both have chance at 0.50 by construction, and that coincidence is the trap: the residual frame's ceiling is 0.958 where the expression frame's is 0.576, so a model scoring 0.569 in one and 0.498 in the other has not improved: it is being measured against a different question.

Frame	Arm	NIR	What it knows
expression (within plate, tier 2, $n = 606$)	noise ceiling	0.576	a real replicate of this condition
	control-copy	0.504	nothing — it reproduces the untreated cell
	linear map	0.500	the control cell, no drug
	model	**0.498**	control cell + drug
	global mean	0.180	nothing
residual (cell-line sets, $n = 570$)	noise ceiling	0.958	a real split-half replicate of this condition
	drug lookup (training-only)	0.941	this drug's signature from other cell lines
	drug lookup, one line	0.859	the same, from a single other cell line
	model	**0.569**	control cell + drug
	scramble, `orth` partner	0.497	the neutral comparator: *empirically* near chance, and selected on the observed outcome
	scramble, `opposite` partner	0.403	an active treatment, not a null
	generic response	0.500	nothing (the zero vector, by construction)

Table 17: Every arm on the calibrated metric, by frame. Chance is 0.50 in both. The residual-frame figures are the rebuilt-target run of §3.11 and §3.12; the held-out subset carries the outcome-selection caveat stated there.

The distance between the model and its neutral scramble is the drug effect this work recovered; the distance between the model and the lookup is what it did not.

4 Limitations and Future Research Directions

4.1 Limitations of scope

Drug coverage. The residual analysis is built on a cache of 620,564 cells covering 106 drugs across 50 cell lines, sampled from Tahoe's shards — roughly a tenth of the atlas's compound set. This is ample to test whether the model uses the drug, and to measure transfer, but it is not a population estimate. Any claim about the *distribution* of drug behaviours should be read against the broader characterisation, which used 6,628 conditions.

Only reproducible conditions. The residual arm retains the 62% of conditions whose residual reproduces across a half-split (cos > 0.2). This is the right choice — training on irreproducible residuals is training on noise — but it means results describe the winnable fraction of the task, not all of it. The filter might be expected to select *toward* context-independent conditions and so bias the transfer coefficient upward. Measured, it does the opposite: the unfiltered estimate is 0.598 against the filtered 0.557. The filtered value is the one quoted, because disattenuation is unstable at the very low reliabilities the filter removes.

The intervals omit generation variance. Every confidence interval in this thesis is a clustered bootstrap that resamples *cell lines* while treating each condition's score as fixed. It therefore captures between-cell-line variance and not the variance introduced by sampling the model's output. With four generations per arm at temperature 0.8 that second source is not negligible, and it is not measured here. An across-seed replication — three seeds, identical in every other respect — would supply the missing component, and the honest form of every interval below would then be a point estimate with a seed spread beside it. [**TODO: run the three-seed replication and report the across-seed spread beside the within-seed interval, stating which sources each captures**]

A swing that looked like generation variance and was not. Two evaluations of the same checkpoint on the same conditions, differing only in that one carried two additional scramble arms and so consumed the sampling stream differently, returned Δ_{scr} on the held-out-drug split as $+0.0195$ $[-0.045, +0.086]$ and $+0.1114$ $[+0.051, +0.173]$. This was originally recorded as evidence of seed instability, and that diagnosis was wrong — or rather, it named the trigger and missed the cause. Both figures are gaps against the `opposite` swap, which Section 3.11 shows is not a null: it contains a term measuring how far the *partner* drug was pushed the wrong way, a quantity that has nothing to do with the target and that changes with which partner happens to be drawn. An arm whose value is partly a nuisance quantity will move when the draw moves. Against the neutral comparator the same control sits at $+0.0097$ $[-0.0655, +0.0848]$. Two mechanisms are involved and both are now addressed: generation was genuinely unseeded — `do_sample` advanced from global torch state, so the two arms of a contrast were drawn from different points of one stream — and each call now seeds its own generator from (condition, arm, batch); and the comparator has been corrected. The general caution above stands regardless; this particular swing is no longer evidence for it.

Pseudobulk resolution. Residuals are condition-level pseudobulks over ≥ 40 treated cells. The single-cell heterogeneity of the response is not modelled in this frame, and the project's own earlier results show why: at single-cell resolution the drug moves a cell less than noise does (SNR ≈ 0.75, median 0 differentially expressed genes at FDR $q < 0.05$).

Cross-plate absolutes. The residual-space scores are computed with comparison sets grouped by cell line rather than by plate. The *difference* Δ_{scr} is leak-immune, because both arms share the same control cell, but the absolute NIR values in that frame are not directly comparable to the within-plate tables elsewhere. [**TODO: rescore the generation-free arms within plate — this costs no GPU time**]

Frame coverage. The drug-specific residual is $\|r\|/\|s\| = 0.786$ of the response, against 0.620 for the generic program and batch, and the two are essentially orthogonal ($0.786^2 + 0.620^2 = 1.002$). So this frame covers 62% of the response *variance* and every result here should be read within that bound — a real but not total share. The remaining 38% is precisely the component the field's absolute-accuracy metrics are dominated by, which is why a drug-agnostic predictor scores so well on them.

4.2 Limitations of the unseen-drug evaluation

Two defects, both of which must be stated rather than papered over. A third — the comparator — was found and corrected rather than merely stated, and Section 3.11 reports the arm against the neutral stratum throughout; the two below survive that correction.

It is not a clean held-out-drug design. The prompt contains `Mechanism: {moa}`, so holding out a drug still hands the model the held-out drug's class label. The correct design is leave-one-*mechanism*-out. The leak is probably small — `moa_lookup` scores 0.529, so the mechanism label carries little about the response — but the design is wrong, and the arm's proper use is as an instrument control rather than as the basis for a positive unseen-drug claim.

It is underpowered by construction. The confidence interval half-width on the `unseen_drug` gap is $\approx \pm 0.075$, against a maximum effect available through the mechanism channel of ≈ 0.033. The arm therefore *cannot* detect the largest effect it could plausibly be measuring. This cuts both ways and the direction matters. As a *control* it is doing the job asked of it: the question there is whether the point estimate sits at zero when the model has no drug knowledge, and +0.0097 does. As *evidence of absence* it carries almost nothing, because an effect of the size the mechanism channel could deliver would sit comfortably inside the interval. The distinction is what licenses using the arm to validate the comparator in Section 3.11 while declining to use it to argue that unseen-drug transfer is zero.

4.3 Directions

The arm this work closed rather than opened. The obvious continuation of Section 3.13 is a channel-conditioned prompt: append protein targets to the instruction, retrain, and evaluate on a leave-one-mechanism-out split. We do not recommend it, and the reason is measured rather than argued. Section 3.14 shows the model reads the drug *name* (+0.081 [+0.059, +0.101]) and does not measurably read the *mechanism* field it is already given (+0.009 [−0.017, +0.038]), in an arm powered to detect an effect the size of the name's. Adding a target field to a prompt whose mechanism field is ignored addresses information availability, and the binding constraint is the readout.

What would be worth building instead is an intervention that makes the drug *causal* rather than merely present — a learned per-drug modulation of the transformation from control to treated, so that the drug parameterises the computation instead of sitting in the residual stream as an input the readout can round away. That is a research architecture rather than a prompt change, and it is outside the scope of what remained here.

Two channel questions that do remain open. Target and mechanism overlap, so whether target adds anything *over* mechanism is unresolved — and it matters, because the prompt already exposes mechanism and the model already ignores it. Separately, the channel gate measures *similarity transfer*: an unseen drug resembling a seen drug with the same target. It does not test a *compositional rule* — that a named target's own pathway moves, which would apply to any drug whose target is named, including one with no similar drug in training. The second is the stronger claim, it does not reduce to a lookup, and it is untested.

Unseen cell lines. A cell line never seen with any drug is a different generalisation axis from the held-out pairings tested here, and one not blocked by the structure gate: it asks whether the control cell's state is used *compositionally* with the drug. The transfer coefficient now scopes it: with the interaction both substantial ($\approx 45\%$) and idiosyncratic, a new cell line offers no consistent direction to exploit, so the achievable target there reduces to recalling the drug's per-drug constant — which a lookup already does at 0.963. The axis is therefore less promising than it appeared before Section 3.9 was measured.

Objectives that were considered and rejected. Recorded so they are not relitigated:

- **Reinforcement learning with a contrastive reward.** Its mechanism — a per-sequence reward, so the token-dilution ratio never enters the gradient — attacks a bottleneck that the residual encoding has already lifted (34.6 $\rightarrow$ 118.2 differing tokens); and its reward optimises similarity to the true residual, which is precisely what `drug_lookup` already achieves at 0.963. Its best case is convergence to the lookup.
- **A differentiable profile-level contrastive loss.** The natural construction, $\mathrm{score}(g) = \sum_i P(\mathrm{token}_i = g)\, w(i)$, presumes gene $\approx$ token, but 86.9% of the 946 panel genes share a first sub-token under this tokeniser (mean 3.254 BPE tokens per gene, only 8 single-token). Under teacher forcing the difficulty compounds: the residual target *is* the drug signature, so the true prefix identifies the drug within a handful of symbols and the gradient vanishes exactly at the drug-blind solution the loss exists to penalise.
- **Loss re-weighting toward differentially expressed genes.** Re-weighted cross-entropy against a single target contains no negatives, so nothing in it forces the output for one drug to differ from another's. Given that the readout is direction-blind (Section 3.6), it addresses token dilution but not the actual defect.

One model size, and the smallest one. Every result here comes from a 1B-parameter backbone. C2S-Scale releases checkpoints up to Gemma-2 at 27B, and we tested none of them, so the most predictable question an examiner can ask — does it work if the model is bigger — has no answer in this thesis.

Two things can be said about what we would expect, and they point in different directions. The bottleneck this work identifies is not capacity. It is that roughly 1% of the training gradient reaches drug identity, because the target tokens are overwhelmingly shared across drugs (Section 3.10); a larger model receives the same diluted signal from the same objective, and the fact that re-encoding the target lifted the effect while leaving capacity untouched is evidence that the constraint binds where we say it does.

The genuinely different argument for scale is not capacity but *prior knowledge*. A 27B model pretrained on far more text plausibly knows, as a fact about the world, that a named compound inhibits a named kinase — knowledge a 1B model may simply lack. That matters here because Section 3.14 finds the model reads the drug *name* but not the *mechanism* field it is already handed, and a model with real pharmacological knowledge is exactly the case where that might change. This is a sharper hypothesis than "bigger is better" and it is testable cheaply with a low-rank adaptation of a mid-sized checkpoint rather than a full fine-tune. We did not run it, and it is the first thing we would run with more time.

The starting checkpoint was never prepared for this task. We fine-tune from `C2S-Scale-Pythia-1b-pt`, which is pretrained on atlas data alone — CELLxGENE and the Human Cell Atlas, no perturbation corpus of any kind — and whose released capabilities do not include perturbation prediction. This cuts two ways and both should be stated. It is why the held-out splits are clean: a drug withheld here was never seen at any stage, so nothing in the pipeline can have memorised it. But it also means the failure we report is a failure of *this* starting point, and a reader may reasonably ask whether a checkpoint already tuned for perturbation prediction would behave differently. We cannot answer that from the public artifacts. The perturbation-specialised model in the same line of work is a Gemma-3 4B checkpoint that was not released, and the available C2S-Scale checkpoints are the pretrained ones. What we can say is narrower and worth separating from the broader claim: every arm in this thesis starts from the same base and differs only in the variable under test, so the *comparisons* between arms are unaffected by this choice even where the absolute level is not.

Generality. Everything here is one backbone on one atlas in one representation. Whether the readout-specificity failure is a property of cell-sentence models specifically, or of autoregressive conditional generation over transcriptomes generally, is open. The mechanistic protocol of Section 3.4 — purified subspace, removal gate, positive control, matched-norm swap — transfers unchanged to any model with an accessible residual stream, and is probably the most reusable artifact of this work.

5 Conclusions

This thesis asked whether a large language model fine-tuned on the largest available single-cell drug-perturbation atlas learns drug-specific biology. It does not — and the interesting content is in the chain of results establishing why, how far the failure can be repaired, and what bounds the repair.

The evaluation had to be fixed first. The metric the field standardises on is exploitable: a zero-information baseline placing every gene at the middle rank scores DE-Δr = 1.000, above both the noise ceiling and the model. Under a Dynamic Range Fraction analysis computed within plate, only NIR — a discrimination metric, in which the generic response cancels — is calibrated. Of the four prediction metrics tested, three are clearly inverted and the fourth is approximately neutral. This is the most portable result here, since it applies to any work scored on DE-Δr-family metrics.

This also settles how the thesis enters the dispute of Section 2.3. Miller et al. [14] argue that reports of deep models failing to beat uninformative baselines reflect miscalibrated evaluation rather than model failure, and on genetic perturbation they support that with calibrated-metric benchmarks. We port their framework, confirm its methodological claim emphatically — metric choice dominates, and only the rank-based metric survives — and then find that on drug perturbation the model is at chance *on the metric their own analysis endorses*. That is the strongest form the sceptical result can take: it cannot be answered with the objection their paper raises. Their conclusion is not overturned so much as bounded — it does not extend from genetic perturbation to this task, and the per-drug-constant finding above suggests why.

The model is drug-blind, and the representation is not the reason. Four independent instruments agree that swapping the drug token changes the output by nothing. Yet drug identity is linearly decodable from the residual stream, rising to 0.88 by layer 16 with context removed, while the purified drug subspace's causal variance share stays flat near 3%. The model computes an increasingly explicit drug representation and does not act on it — a dissociation between what is encoded and what is used.

How far that can be pushed toward a mechanism is limited, and the limit should be stated plainly. Two earlier attempts to localise the failure by activation swapping were retracted after their matched controls were found confounded, and what replaces them (Section 3.6) is a small, replicated, single-layer effect present in one training arm and absent in another. That supports the modest reading — the generation readout consults the drug representation at very low gain — and does not support a claim about privileged subspaces or global workspaces, which is offered here as an interpretive frame rather than a result.

The bottleneck appears to be the tokenisation, and it is repairable. Because a cell sentence ranks genes by expression, the targets for two different drugs in the same context are nearly the same string: 82.7% of the top-200 genes are shared, so only about a sixth of the target distinguishes one drug from another. Since the training loss weights every token position equally, the natural inference is that very little of the gradient can be carrying drug identity — but that inference is a *hypothesis* about the optimisation, not a measurement of it. What is measured here is the overlap of the target gene sets; no token-level loss or gradient attribution was computed, and the claim should be read at that strength. What follows is that the hypothesis makes correct predictions, which is the evidence for it. It explains a uniform pattern of failure: an entire ε-ladder of target constructions, from a single state-matched treated cell to a full pseudobulk collapse, stays in the near-null regime, because all of them change *what the target is* and none change *what the tokens encode*. (One arm, the optimal-transport target, does show a weak effect once a sharper stratified test is applied — +0.0263 [+0.0069, +0.0476] — but roughly five-fold below what re-encoding achieves, which is the distinction that matters.) Re-encoding the target as the drug-specific residual raises the differing fraction to 59% and produces the first non-null drug effect in this project, with a monotone gradient across swap dissimilarity and a matched control that stays null.

What that drug use amounts to: a main effect, not an interaction. An earlier form of this paragraph claimed the drug use generalises, on a holdout keyed at the (drug, cell line) level where pairings never seen in training scored +0.1002 [+0.0661, +0.1368] and held-out *drugs* came out null. Two corrections have since removed that claim. The targets were rebuilt so the generic is fitted after the split rather than over the holdout, and the comparator was corrected: the scramble arm the gap was measured against is not a null but a different drug's learned signature, chosen to point away from the truth. The held-out-drug arm is what exposed it, by returning a positive result on a split where a positive result is impossible.

Measured against the neutral comparator on the rebuilt targets, trained pairings score +0.1457 and held-out pairings +0.0449 $[-0.0074, +0.0972]$ — so there *is* a memorisation premium (+0.1008, cluster-robust $p = 0.015$), and cross-context transfer is not established. It is not shown to be zero either: the interval reaches +0.097, and at $n = 250$ that is a limit of power. What does survive, on two tests that use no comparator at all and so cannot be reached by this defect, is that the model emits the signature of the drug it is named (+0.374 on held-out pairings against +0.390 on trained ones) while scoring at chance when ranked against other drugs in the same cell line. The map exists and is coarse: the model learned the drug main effect and not the drug $\times$ cell-line interaction — which is also why the lookup below beats it, since the lookup *is* that main effect, measured directly.

But a lookup table wins — and it is not the ceiling either. A per-drug mean residual measured in other cell lines scores 0.963 against a 0.968 replicate ceiling; the model reaches 0.639. The natural reading of those three numbers is that the drug's response is essentially a per-drug constant, that a conditional model therefore has almost nothing to condition on, and that the correct formulation of the task is retrieval rather than generation. We initially drew exactly that conclusion, and it is wrong.

It is wrong because NIR compresses near the top of its range — the same class of failure this thesis opens by documenting, arriving a second time in our own preferred instrument, at the ceiling instead of the floor. Measured in cosine space with measurement noise divided out, the transfer coefficient is $\mathcal{T}$ =0.557 [0.513,0.601] against a simulated no-interaction null of 1.001, with a structure-matched negative control at +0.000. Roughly 45% of the drug-specific signal is a drug×cell-line interaction. The per-drug constant is what the lookup captures and what the model gropes toward; the interaction is reached by neither, and by nothing else in the literature we are aware of.

So the honest statement of what the model does is sharper, and less flattering, than "it misses the interaction". It fails to recover the *main effect* that a numpy dictionary obtains for free, and the interaction is a further prize that no method here — ours least of all — comes near. Two different failures, and collapsing them undersells both.

That conclusion connects this work to an assumption the field has carried since the Connectivity Map: that a compound induces a characteristic signature stable enough across cellular contexts for signature-similarity querying to be meaningful. This thesis measures that assumption directly, with a calibrated metric and a proper replicate ceiling, at single-cell resolution — and the modern finding that a per-perturbation mean is hard to beat may be less a warning about weak baselines than a restatement of it.

It also places this work as the single-cell, generative instance of DrEval's critique [13]. DrEval audits drug *sensitivity* prediction and finds that deep models barely outperform a naive additive predictor of mean drug and cell-line effects. Our `drug_lookup` is that predictor's drug term, and the finding replicates on a 946-dimensional transcriptomic response: 0.963 against a 0.968 replicate ceiling, versus 0.639 for the model. Three of DrEval's six pathologies — pseudoreplication, biased evaluation metrics, and missing ablation studies — turn out to structure the entire methodology of this thesis, which arrived at clustered bootstraps over cell lines, a discrimination metric immune to between-drug variance, and a token-level scramble ablation independently. That convergence is worth recording: the controls a critical benchmark demands are the same controls a single negative result needs in order to mean anything.

Where the opportunity actually is. Two further measurements locate it precisely, and between them they redirect what follows from this work.

The drug×cell-line interaction is large — roughly 45% of the drug-specific variance —

and we could not find structure in it. No cell line modulates different drugs in a consistent direction (excess within-line consistency -0.007, and at plate scope an interval excluding anything above $+0.002$), and a direct test of whether mechanism-matched drugs share an interaction with a given cell line returned nothing detectable (permutation $p \approx 0.15$–0.31).

Those are two negative results and they do not license the conclusion that the interaction is irreducible. The second test bounds its own power severely: the interaction's split-half reliability in this atlas is 0.195, so κ is barely estimable, and same-mechanism compounds are so heavily co-plated that only 3–7% of matched pairs survive a different-plate requirement. The honest statement is that the structure of the interaction is *unresolved here*, and that resolving it needs more cells per condition or a plating design that crosses mechanism with plate — neither of which is an analysis choice. What can be said without qualification is that a lookup scoring 0.963 against a 0.968 ceiling is already close to what a per-drug constant can achieve, so on seen drugs nothing we built beats retrieval.

But a lookup requires the drug to have been seen. For drugs that have not been, protein-target and mechanism annotations score above a *count-matched* null (+0.084 [+0.059, +0.110] and +0.078 [+0.056, +0.103]), while chemical structure does not. That null is matched on the number of partners but not on plate: drugs sharing a mechanism are largely laid out on the same plate, drugs drawn at random are not, and the residuals being averaged retain plate structure. The arm and its null therefore differ in two respects where the estimand allows one, so an unknown share of these gaps is plate rather than annotation. A plate-matched re-measurement is pending and these two numbers should not be read as established until it lands. This overturns a pre-registration made earlier in this project, which inferred that unseen drugs were unreachable from a gate that tested chemical *structure* — the channel our own drug-knowledge specification ranked lowest — and never read the target annotations sitting in the atlas's own metadata.

So the thesis does not end in closure. It ends with the opportunity located: on seen drugs a generative model has nothing to add over retrieval, because the residual part is unpredictable; on unseen drugs, where retrieval is impossible by construction, drug-side knowledge works and a model can do something a dictionary cannot. That is also the setting the benchmark literature calls Leave-Drugs-Out, and the one that matters for drug design.

A note on method. The substantive results here rest on controls that were added in response to specific objections: plate-matched comparison sets after a zero-information baseline more than doubled its score when the plate was free to vary; a swap-distance sweep after it was pointed out that scrambling to a similar drug is a weak test; a removal gate before trusting any ablation null; matched-norm injection before reading a swap; clustered bootstraps over cell lines rather than conditions; and a pre-registered interpretation written before the decisive result, whose middle branch is the one that fired. Several claims were retracted along the way and are struck rather than deleted. In a literature where the central problem is that uninformative baselines score well, the controls are not an appendix to the result — they are most of it.

References

[1] Jesse Zhang et al. "Tahoe-100M: A Giga-Scale Single-Cell Perturbation Atlas for Context-Dependent Gene Function and Cellular Modeling". In: *bioRxiv* (2025). Preprint, posted February 2025 (v3, 10 May 2025); not peer reviewed. Tahoe Therapeutics, formerly Vevo Therapeutics. Dataset: `https://huggingface.co/datasets/tahoebio/Tahoe-100M`. DOI: `10.1101/2025.02.20.639398`.

[2] Daniel Levine et al. "Cell2Sentence: Teaching Large Language Models the Language of Biology". In: *Proceedings of the 41st International Conference on Machine Learning (ICML)*. Vol. 235. Proceedings of Machine Learning Research. Preprint: bioRxiv 2023.09.11.557287. PMLR, 2024, pp. 27299–27325.

[3] Syed Asad Rizvi et al. "Scaling Large Language Models for Next-Generation Single-Cell Analysis". In: *bioRxiv* (2025). Preprint, v1 posted 17 April 2025; author list above is v4, posted 1 January 2026. Not peer reviewed. DOI: `10.1101/2025.04.14.648850`.

[4] Haotian Cui et al. "scGPT: toward building a foundation model for single-cell multi-omics using generative AI". In: *Nature Methods* 21.8 (2024), pp. 1470–1480. DOI: `10.1038/s41592-024-02201-0`.

[5] Christina V. Theodoris et al. "Transfer learning enables predictions in network biology". In: *Nature* 618.7965 (2023), pp. 616–624. DOI: `10.1038/s41586-023-06139-9`.

[6] Charlotte Bunne et al. "How to build the virtual cell with artificial intelligence: Priorities and opportunities". In: *Cell* 187.25 (2024), pp. 7045–7063. DOI: `10.1016/j.cell.2024.11.015`.

[7] Stella Biderman et al. "Pythia: A Suite for Analyzing Large Language Models Across Training and Scaling". In: *Proceedings of the 40th International Conference on Machine Learning*. Vol. 202. Proceedings of Machine Learning Research. PMLR, 2023, pp. 2397–2430. URL: `https://proceedings.mlr.press/v202/biderman23a.html`.

[8] Constantin Ahlmann-Eltze, Wolfgang Huber, and Simon Anders. "Deep-learning-based gene perturbation effect prediction does not yet outperform simple linear baselines". In: *Nature Methods* 22.8 (2025), pp. 1657–1661. DOI: `10.1038/s41592-025-02772-6`.

[9] Gerold Csendes et al. "Benchmarking foundation cell models for post-perturbation RNA-seq prediction". In: *BMC Genomics* 26.1 (2025), p. 393. DOI: `10.1186/s12864-025-11600-2`.

[10] Ihab Bendidi et al. "Benchmarking transcriptomics foundation models for perturbation analysis: one PCA still rules them all". In: *arXiv preprint arXiv:2410.13956* (2024).

[11] Eric Kernfeld et al. "A comparison of computational methods for expression forecasting". In: *Genome Biology* 26.1 (2025). Preprint: bioRxiv 2023.07.28.551039, titled "A systematic comparison of computational methods for expression forecasting", p. 388. DOI: 10.1186/s13059-025-03840-y.

[12] Tony Hauptmann and Stefan Kramer. "A fair experimental comparison of neural network architectures for latent representations of multi-omics for drug response prediction". In: *BMC Bioinformatics* 24.1 (2023), p. 45. DOI: 10.1186/s12859-023-05166-7.

[13] Judith Bernett et al. "From Hype to Health Check: Critical Evaluation of Drug Response Prediction Models with DrEval". In: *bioRxiv* (2025). Preprint, posted 29 May 2025. DOI: 10.1101/2025.05.26.655288.

[14] Henry E. Miller et al. "Deep Learning-Based Genetic Perturbation Models Do Outperform Uninformative Baselines on Well-Calibrated Metrics". In: *bioRxiv* (2025). Preprint, posted 21 October 2025. Shift Bioscience Ltd. DOI: 10.1101/2025.10.20.683304.

[15] Gabriel M. Mejia et al. "Diversity by Design: Addressing Mode Collapse Improves scRNA-seq Perturbation Modeling on Well-Calibrated Metrics". In: *arXiv preprint arXiv:2506.22641* (2025).

[16] Yan Wu et al. "PerturBench: Benchmarking Machine Learning Models for Cellular Perturbation Analysis". In: *arXiv preprint arXiv:2408.10609* (2024).

[17] Stuart H. Hurlbert. "Pseudoreplication and the Design of Ecological Field Experiments". In: *Ecological Monographs* 54.2 (1984), pp. 187–211. DOI: 10.2307/1942661.

[18] Alexander Partin et al. "Deep learning methods for drug response prediction in cancer: Predominant and emerging trends". In: *Frontiers in Medicine* 10 (2023). DOI: 10.3389/fmed.2023.1086097.

[19] Justin Lamb et al. "The Connectivity Map: using gene-expression signatures to connect small molecules, genes, and disease". In: *Science* 313.5795 (2006), pp. 1929–1935. DOI: 10.1126/science.1132939.

[20] Aravind Subramanian et al. "A Next Generation Connectivity Map: L1000 Platform and the First 1,000,000 Profiles". In: *Cell* 171.6 (2017), 1437–1452.e17. DOI: 10.1016/j.cell.2017.10.049.

[21] Yanai Elazar et al. "Amnesic Probing: Behavioral Explanation with Amnesic Counterfactuals". In: *Transactions of the Association for Computational Linguistics* 9 (2021), pp. 160–175. DOI: 10.1162/tacl_a_00359.

[22] Wes Gurnee, Nicholas Sofroniew, Adam Pearce, et al. "Verbalizable Representations Form a Global Workspace in Language Models". In: *Transformer Circuits Thread* (2026). Published 6 July 2026. https://transformer-circuits.pub/2026/workspace/index.html.

[23] Nelson Elhage et al. "Toy Models of Superposition". In: *Transformer Circuits Thread* (2022). arXiv:2209.10652.

[24] Marco Cuturi. "Sinkhorn Distances: Lightspeed Computation of Optimal Transport". In: *Advances in Neural Information Processing Systems 26 (NIPS 2013)*. Ed. by C.J. Burges et al. Vol. 26. Curran Associates, Inc., 2013, pp. 2292–2300. URL: https://proceedings.neurips.cc/paper_files/paper/2013/file/af21d0c97db2e27e13572cbf59eb343d-Paper.pdf.

[25] Geoffrey Schiebinger et al. "Optimal-Transport Analysis of Single-Cell Gene Expression Identifies Developmental Trajectories in Reprogramming". In: *Cell* 176.4 (2019), 928–943.e22. DOI: 10.1016/j.cell.2019.01.006.

[26] Romain Lopez et al. "Deep generative modeling for single-cell transcriptomics". In: *Nature Methods* 15.12 (2018), pp. 1053–1058. DOI: 10.1038/s41592-018-0229-2.

A Appendix

A.1 Estimator validation against planted ground truth

Three of the measurements in this thesis rest on estimators built for the purpose rather than taken off the shelf. Each was validated by planting a known answer and checking recovery — in both directions, since an estimator that only ever returns the answer one hopes for is not evidence.

Estimator	Planted truth	Recovered	
transfer coefficient $\mathcal{T}$	$\mathcal{T} = 1.00$ (no interaction)	0.999	
	$\mathcal{T} = 0.70$	0.703	
	$\mathcal{T} = 0.40$	0.404	
channel gate	a channel that genuinely predicts	+0.397 (sd 0.029)	over 24 worlds
	a channel carrying nothing	−0.016 (sd 0.091)	over 24 worlds
κ-structure	a consistent per-line modulation	excess +1.085	
	an idiosyncratic interaction	excess +0.001	

Table 18: All three recover planted answers in both directions.

Two of these validations changed the estimators rather than merely endorsing them. The channel gate's count-matched null originally drew its random partners from a pool *excluding* the channel's own selections, which depletes the null of exactly the partners the channel chose and anti-correlates the two arms. And the useless-channel gap has a standard deviation of roughly 0.09 on a single small world, so the original validation — asserting that one draw fell within 0.10 — tested luck rather than correctness; it now averages 24 independent worlds. That variance is also why the reported channel results are gated on a clustered interval rather than on a point estimate.

A.2 Absent-gene convention

A predicted cell sentence lists only expressed genes, so some panel genes are unranked and a convention is required. Two were carried throughout: `worst` assigns absent genes the last rank G, `francesca` a fixed middle rank $G/2$. The choice does not matter — spike-in DE-Δr reads 0.952 under the first and 0.954 under the second, and no ceiling or baseline in this thesis moves by more than 0.01 between them. All figures in the body are quoted under `worst`.

A.3 Swap-distance sweep

The stratified scramble of Section 3.3 was preceded by a continuous version: Δ_{scr} measured as a function of the response distance between the real and the swapped-in drug, binned into quintiles.

Bin (swap distance)	single-cell model	optimal-transport model
0 — nearest (2.75–4.03)	−0.046	−0.001
1 (4.03–4.60)	+0.060	+0.031
2 (4.60–4.93)	−0.006	+0.017
3 (4.93–5.32)	−0.004	+0.043
4 — farthest (5.32–6.25)	−0.019	−0.005
trend, corr(distance, Δ_{scr})	+0.051	+0.012

Table 19: No bin in either model has an interval excluding zero, and the trend correlations are ≈ 0. 200 (drug, swap) pairs over 20 cell lines, $n = 40$ per bin.

Two features make this an argument rather than five nulls. The single borderline value sits in a *middle* bin, whereas a genuine dissimilarity effect would be monotone and peak in the far bin. And the comparison is paired: the model's own NIR drifts across bins from 0.497 to 0.326, because drugs that are far from everything populate the far bins, but Δ_{scr} uses the same drug on both arms so that drift cancels.

The intervals are wide — $n = 40$ per bin gives half-widths of 0.06–0.10 — so this sweep excludes a large effect rather than establishing a null, which is why the stratified scramble replaced it in the body.

A.4 Training configuration

Every arm was cold-started from the same base checkpoint with identical hyperparameters, so that the target or objective is the only variable. One exception is material and is stated wherever the arm is quoted.

Arm	Target	Training	Note
single cell	the observed treated cell	1 epoch	
consensus	group pseudobulk	$\approx$ 60% **of an epoch**	tier-1 NIR unscoreable
optimal transport (T2)	barycentric, ε=0.5	1 epoch, 23,842 steps	
residual	signed residual signature	1 epoch, 15,340 steps	
residual (holdout)	as above, 738 withheld	1 epoch	

Table 20: Learning rate 10^{-5}, batch size 1, gradient accumulation 16, maximum length 8192, bf16, weight decay 0.01, warmup ratio 0.03 throughout. The consensus arm stopped early on a flat loss; since it is one endpoint of the ε-ladder, the ladder's negative rests on the other two.

A.5 The identity test: failed designs and remaining guards

The substitution test of Section 3.6 is the fourth design. The three that failed are recorded here with their algebra, because each produced results this thesis previously quoted as findings, and the constraints they teach are the surviving design.

Failed design one: mismatched geometry. The first test overwrote drug A's code with drug B's, inside a $\leq$ 10-dimensional purified subspace, and compared the resulting KL against a random vector of identical norm drawn in the *full* 2048-dimensional stream. Roughly 99.5% of the control's energy then lay in directions the intervention never touched, including the cell-line directions measured at ≈ 0.6 variance share. "The swap moves the output less than noise" is exactly what that geometry predicts, whatever the readout does.

Failed design two: ablate-then-add. The second drew the control inside the subspace, which looked like a repair and was not. The hook ablates before adding, so what is delivered is (added − removed), and an uncentred class mean shares its global-mean component with the thing removed. The substitution arm was then close to a restoration and the control arm a real displacement. Simulated with no model present, the control delivers 5.3× the perturbation in 100% of draws, with the predicted gap $2\langle \mu_B, u \rangle = 9.179$ against an observed 9.162, where μ_B is the injected class mean and u the cell's own component in the slab. The retracted result was geometry, not biology.

Failed design three: the normaliser. The third injected the displacement and scored $\Delta \log P(g_B)$ against a norm-matched random direction — and was confounded the opposite way. A random direction inside the slab is off the class-mean manifold: it inflates the logsumexp normaliser (measured +2.75 nats) where the on-manifold displacement does not, so every target is depressed under the control arm whether or not it carries identity. With the scored target stripped of all identity information, 36% of the effect still stood. Norm-matched is not disruption-matched. The same argument retires the simplest test of all, "how far did the output move": distance moved is exactly what norm-matching equalises, so that test has no power in precisely the world where the readout does read the drug.

Three further guards from the surviving design are engineering rather than inference and are recorded here. A *drug-proxy screen* refuses any context label that is a near-bijection with drug — it caught the metadata field `dose_float` holding a sample identifier in the residual builders, which, had it been orthogonalised against, would have deleted the drug axis by construction. An *injected-norm diagnostic* reports the displacement against the cell's centred slab component (0.69–1.40: the injection is the size of the thing it displaces, so a null cannot be blamed on a whisper). And a *selftest with teeth* runs three planted worlds — drug inert, drug read, and drug inert behind a norm-sensitive readout — through the actual headline code path; it separates them (−0.042, +1.520, −0.010), and it failed four times during development, each failure a real defect that the guard list now covers.

A.6 Calibration self-test

The calibration framework of Section 3.2 is itself checked, since a metric audit whose instrument is untested proves nothing. Beyond deterministic unit checks — a perfect prediction must reach the metric's maximum, a zero shift and a flat truth must return no score — the self-test asserts the exploit under a *second, independent* zero-information baseline: a leave-one-out mean scores DE-$\Delta r = 1.000$ while scoring NIR $= 0.000$. Two unrelated constructions both saturating the standard metric, and both scoring zero on the discrimination metric, establishes the exploit as a property of the metric rather than of any one baseline.